

MAS2803/PHY2033

Fluid Dynamics

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Semester 2, Academic Year 2019/2020

Preface

The notes are structured as follows: in Section 1 details and the timetable of the course are outlined. Section 2 aims at an introduction of fluids: what are fluids and why we would want to study them. An introduction to physical dimensions and units and a revision of Cartesian and cylindrical polar coordinates systems is also provided. Finally the concept of density and pressure are revised and treated within the so-called continuum approximation. Section 3 is aimed at the study of how fluids flow. The concept of flow velocity is introduced and a first classification of flows is provided according to the fundamental properties of flows. Streamlines and pathlines are defined and flows are studied from two different perspectives: Eulerian and Lagrangian. The material derivative represents an important mathematical tool which allows us to study the fluid acceleration. Vortices and connections with circulations represent the last part of the section, which contains several remarkable physical examples at its end. Section 4 is devoted to the study of static fluids. This section is fundamental for understanding forces that act on fluids and how fluids respond to forces. A study on pressure and pressure forces on fluid surfaces and fluid volumes is presented. An introduction to body forces is provided and the case of gravity is studied in detail. The remainder of the section is focused on the study of static of incompressible fluids. Section 5 introduces the concept of mass flux. The conservation of mass and its consequences in the study of incompressible flows are outlined. The section introduces the fundamental equations and theorems of the course: the continuity equation, the Euler equation, its reduction to incompressible fluids, and the vorticity equation. The final part of the section is based on Bernoulli's theorem and its consequences and applications. The core part of the notes concludes with Section 6, where irrotational and incompressible flows are studied. Firstly the concepts of velocity potential and potential flow are introduced. Laplace's equation for incompressible and irrotational fluids is derived and studied. The section ends with the study of the fluid dynamics of specific and important examples. In the appendices Section A, several calculations and theorems useful for understanding the results discussed in the core sections are shown in detail.

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1 Preliminaries

1.1 Module details

Overview

This course will introduce what a fluid is and the fundamental concepts and governing equations of fluid mechanics. The main focus will be on fluids which have no viscosity (friction). Even with such a simplifying assumption, it is not possible to write down a general solution of the governing equations. However, in certain fundamental scenarios we can deduce the properties of the flow. These results will give useful insight into fluid motion in nature and engineering situations.

We will use ideas from MAS1902/PHY1020 Dynamics (position, velocity and acceleration vectors, forces, and Newton's second law) and MAS2801/PHY2026, Vector Calculus (differential operators and integral theorems) .

Contact

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Timetable

- Monday 09:00-10:00, Curtis Auditorium, Herschel Building
- Tuesday 15:00-17:00, Alec Campbell Lecture Theatre - Lecture Theatre 1 G.04, Bedson

Sessions

- *Lectures*: New material will be introduced and discussed, following the Lecture Notes provided but with additional comments and asides. We will work through many important examples on the whiteboard (the solutions to these will **not** be available electronically).
- *Problem Classes*: A focussed session working through further exercises.
- *Drop-In Sessions*: Non-compulsory sessions where you may work through example questions or assignment questions, with the module leader at hand to provide assistance. You can also raise any problems you are having with course material.
- *Revision Lectures*: We will work through a sample exam paper.

Assessment

- 2 Assignments (2.5% each)
 - Deadline **Thursday** 15:00 in weeks 5 & 11
- Test (10%)
 - 16:00 **Tuesday** 10 March (week 7)
- Exam (85%)
 - 2 hour exam

Late work policy

- It is not possible to extend submission deadlines for coursework in this module and no late work can be accepted. The module contains one or more in-course tests that cannot be rescheduled. For details of the policy (including procedures in the event of illness etc.) please look at the School web site (<http://www.ncl.ac.uk/maths/students/teaching/homework/>).

Course Material

- These lecture notes are a version of the material presented during lectures, but should not be taken as a replacement for all that is done in the lectures. During the lectures I will present various asides and comments that do not feature in the lecture notes, and go through the examples on the whiteboard. You are recommended to bring the printed Lecture Notes to all sessions. Important definitions or results are highlighted in grey boxes.
- All other course material (e.g. lecture notes, hand-outs, assignments and model assignment solutions) will be available on the course webpage <http://abag.wikidot.com/teaching>.
- All lectures will be Recapped and posted on Blackboard.
- Supplementary Exercises sheets will be provided for each Chapter, and the solutions made available.

Resources

- *Elementary Fluid Dynamics*, David Atcheson, Clarendon Press (1990)
- *Fluid Mechanics*, Pijush K. Kundu & Ira M. Cohen (2008)
- FYFD Fluid Dynamics Blog <http://fuckyeahfluidynamics.tumblr.com/>

1.2 Timetable

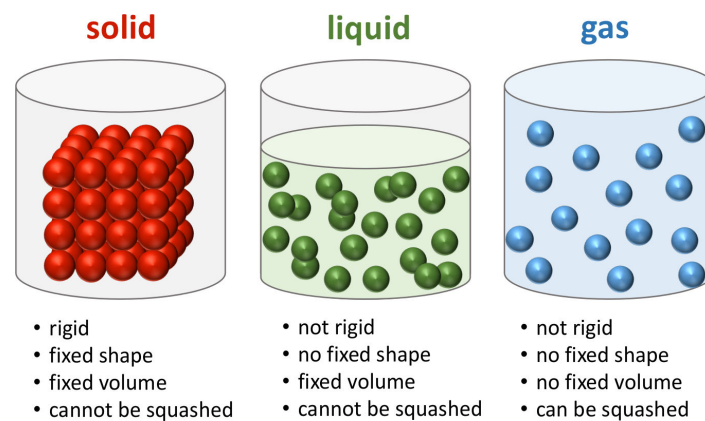
WEEK (BEGINN.)	DAY	TIME	SESSION	LOCATION
1 (27 Jan)	Mon	0900	No session	
	Tue	1500	Lecture 1 & 2	LT1 G.04, Bedson Building
2 (3 Feb)	Mon	0900	Lecture 3	CA, Herschel Building
	Tue	1500	Lecture 4 & 5	LT1 G.04, Bedson Building
3 (10 Feb)	Mon	0900	Drop-in	CA, Herschel Building
	Tue	0900	Lecture 6 & 7	LT1 G.04, Bedson Building Assignment 1 released
4 (17 Feb)	Mon	0900	Problem Class 1	CA, Herschel Building
	Tue	1500	Lecture 8 & 9	LT1 G.04, Bedson Building
5 (24 Feb)	Mon	0900	Drop-in	CA, Herschel Building
	Tue	1500	Lecture 10 & 11	LT1 G.04, Bedson Building
	Thu	1500		Assignment 1 due
6 (2 March)	Mon	0900	Problem Class 2	CA, Herschel Building
	Fri	1500	Lecture 12 & 13	LT1 G.04, Bedson Building
7 (9 March)	Mon	0900	Drop-in	CA, Herschel Building
	Tue	1600	Test	See your personal timetable
8 (16 March)	Mon	0900	Lecture 14	CA, Herschel Building
	Tue	1500	Lecture 15 & 16	LT1 G.04, Bedson Building
9 (23 March)	Mon	0900	Problem Class 3	CA, Herschel Building
	Tue	1500	Lecture 17 & 18	LT1 G.04, Bedson Building Assignment 2 released
EASTER				
10 (27 April)	Mon	0900	Drop-In	CA, Herschel Building
	Tue	1500	Lecture 19 & 20	LT1 G.04, Bedson Building
11 (4 May)	Mon	0900	No lecture	Bank Holiday
	Fri	1500	Lecture 21 & 22	LT1 G.04, Bedson Building
	Thu	1500		Assignment 2 due
12 (11 May)	Mon	0900	Drop-in	CA, Herschel Building
	Tue	1500	Revision Lecture 1 & 2	LT1 G.04, Bedson Building

2 Introduction

2.1 States of matter

We are familiar with the three states of matter: *solid*, *liquid* and *gas*. A solid is rigid and resistant to being deformed; a force may cause some displacement of the particles within but the solid will not move indefinitely. Liquids and gases have no fixed shape, cannot resist being deformed if acted on by some force, and can be made to flow. They are collectively termed *fluids* and the study of their flow is *fluid dynamics*.

There can be vast differences in physical properties of fluids. Liquids are dense (water, for example, has a density of 1000 kg m^{-3}) and hard to compress, tending to take up a fixed volume. The study of fluid dynamics specifically in liquids is *hydrodynamics*. Gases are light (air, for example, has a density of around 1 kg m^{-3}), tend to fill up the container they are in, and are easy to compress. The study of fluid dynamics specifically in gases is *aerodynamics*. But there are many common behaviours, as we will see.



The three main states of matter and their defining properties.

2.2 Fluids

Fluids are all around us, from the air we breathe and the water we drink, to the petrol in our cars and the atmosphere circling above us. Understanding how fluids move is central to the natural world. Fluid motion in the body, e.g. respiration and blood flow, is vital to how we stay alive. The fluid motion of the atmosphere makes our planet suitable for life, while the flow of the interstellar medium gave rise to today's planets and stars. Fluid dynamics is also important in industry.

The motion of fluids can occur across a huge range of lengthscales, from the internal running of biological cells at the nanometre-scale (1 nanometre (nm) is 10^{-9}m , about as big as a DNA molecule) up to the megaparsecs of galactic structures (1 megaparsec is around 3 million light-years).

It is not surprising that fluid dynamics is one of the most important areas of science and engineering. The ability to predict fluid dynamics allows us to forecast tomorrow's weather, reduce turbulence on flights, and make more efficient vehicles and power stations.

Fluid dynamics also extends beyond conventional liquids and gases. We talk about the "flow" of electricity, heat, crowds, traffic and grain. Some solid-like substances can flow if we wait long enough: for example, a drop of tar appears as a solid over days and weeks, but is seen to flow over decades (www.youtube.com/watch?v=k7jXjn7mIao).



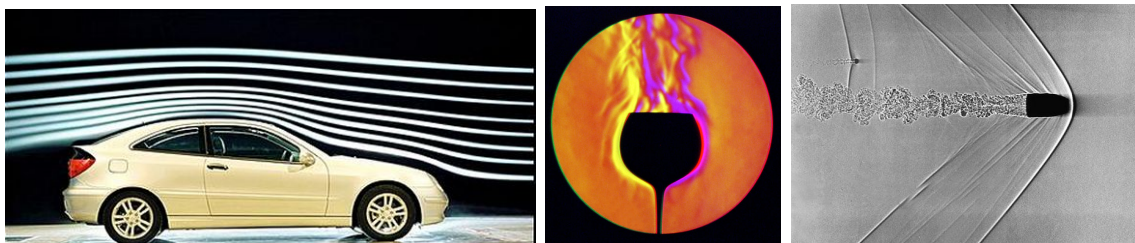
Examples of fluid dynamics. Clockwise from top-left: i) Striking cloud formation. ii) Plane landing in a cross-wind. iii) Beer. iv) Aerodynamics of a cyclist. v) Blood flow in the heart. vi) The Milky Way galaxy. vii) Air flow past wind mills. viii) An autumn breeze kicks up some leaves.



Fluid dynamics in systems which are neither liquid or gas. From left: i) A grain pile. ii) A land slide. iii) Flow of electricity. iv) The tar drop experiment, Trinity College Dublin.

2.3 Flow visualization

We cannot appreciate with the naked eye the fluid motion going on around us. What we might sense as the breeze on our skin or see as a leaf swirling in the air or the steam trail above our coffee cup, is only a tiny glimpse of the wider flow. This has motivated the clever use of “tracers” and dyes, sophisticated photography and mock-ups, to capture and understand fluid dynamics. We will meet some of these images along the way.



Clever tricks can help to visualize fluid motion. From left: i) Smoke streams past a car in a wind tunnel. ii) Convection of air around a warm glass. iii) A bullet through air.

2.4 Dimensions and units

What are dimensions and units?

We will often apply our ideas to physical settings. To help quote correct and meaningful results we will establish some rules about physical dimensions and units.

A **physical dimension** is a fundamental physical variable (without any specification of value), for example, length and time. We quote the value of a physical dimensions in **units**, for example, metres and seconds.

Some quantities, such as the number of seats in this room or the number of leaves on a tree, are just numbers, and have no dimensions; these are *dimensionless quantities*.

By default we employ S.I. units: kilograms (kg) for mass, metres (m) for length and seconds (s) for time. However, we do not *have* to use S.I. units; we can choose units which are most convenient (when asked your age we answer in units of years, not seconds), but we must state the units used otherwise the answer is incomplete*.

We denote units using square brackets. For example, $[x] = \text{m}$ means “the units of x are metres”.

From the basic dimensions of **mass**, **length** and **time** we can construct all of the other dimensions used in this module, for example, speed, acceleration and density. The same is true for their units.

For example, speed is length divided by time, so the dimension of speed is simply *length/time*. From this we can derive the units for speed algebraically as,

$$[\text{speed}] = \frac{[\text{length}]}{[\text{time}]} = \frac{\text{m}}{\text{s}}.$$

Acceleration is speed divided by time, so the dimension of acceleration is *speed/time* = *length/time*² and the units are,

$$[\text{acceleration}] = \frac{[\text{length}]}{[\text{time}]^2} = \frac{\text{m}}{\text{s}^2}.$$

Density is mass divided by volume, so the dimension is *mass/volume* = *mass/length*³ and the units are,

$$[\text{density}] = \frac{[\text{mass}]}{[\text{length}]^3} = \frac{\text{kg}}{\text{m}^3}.$$



EXAMPLE 1: You may be familiar with quoting force in units of Newtons, pressure (force divided by area) in terms of Pascals, and energy in terms of Joules, but what are these units in terms of the kilograms, metres and seconds?

*This is true when dealing with some physical situation, although often in mathematics we choose to ignore units; it should be obvious in a given question whether units are involved/required or not.

Dimensional analysis

Any physically meaningful equation should have the same dimensions/units on the left and right sides.. Saying that seven kilograms is equal to three hours is clearly bonkers. Saying that the position of an object is given by $x = at$ where a is its acceleration and t time is also bonkers, since the left side has overall dimensions of length but the right side has overall dimensions of length/time (a dimensionally correct equation would be $x = vt$, where v is the object's speed). This is useful to check whether equations you obtain are correct (at least in terms of their dimensions).

Converting between units

Quantities can be stated in a multitude of different units, either for historical reasons (pints instead of litres) or convenience (age in years versus age in seconds). Converting between two different units can be done algebraically, providing we know how the units are related. Imagine we want to write the area of 3 square-metres in square-nanometres. We proceed as follows:

$$3 \text{ m}^2 = 3 \left(\frac{1 \text{ m}}{1 \text{ nm}} \cdot 1 \text{ nm} \right)^2 = 3 \left(\frac{1 \text{ m}}{1 \text{ nm}} \right)^2 \text{ nm}^2 = 3 (10^9)^2 \text{ nm}^2 = 3 \times 10^{18} \text{ nm}^2.$$

We multiply the base unit, m, top and bottom, by the required unit, nm, and then manipulate taking into account the conversion factor, $1\text{m}/1\text{nm}=10^9$. This approach allows us to convert between any units.



EXAMPLE 2: A substance has a density of 120 kg/m^3 . Convert this to g/mm^3 .
A group of trend-setting scientists prefer to quote their densities in terms of baboons per jam jar, where 1 baboon is defined as having a mass of 14 kg and a jam jar has volume 0.001 m^3 . Convert to this unit.

2.5 Coordinate systems

We now revise the two coordinate systems we will use throughout this module - Cartesian coordinates and cylindrical polar coordinates.

Cartesian coordinates

A point is specified by a position vector,

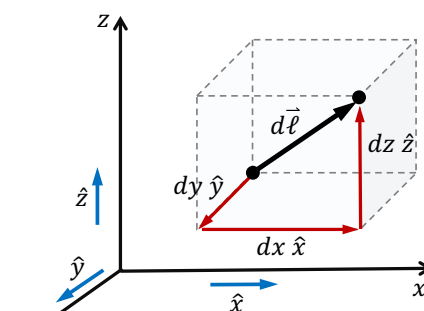
$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z} \equiv (x, y, z),$$

where $\hat{x}$, $\hat{y}$ and $\hat{z}$ are the unit vectors.

The line element vector[†] (the small displacement vector between two points) is,

$$d\vec{\ell} = dx \hat{x} + dy \hat{y} + dz \hat{z} \equiv (dx, dy, dz).$$

The surface element for a surface of constant z is $dS = dx dy$ (and similarly for surfaces in other planes). The volume element is $dV = dx dy dz$.



[†]Sometimes called the *differential displacement vector*.

Cylindrical polar coordinates

A point is specified by position vector,

$$\vec{x} = r \hat{r} + \theta \hat{\theta} + z \hat{z} \equiv (r, \theta, z),$$

where r is its radius from the z -axis, θ is its azimuthal angle (subtended from the positive x -axis) and z is its height along the z -axis. The unit vectors are $\hat{r}$ (pointing radially outwards), $\hat{\theta}$ (pointing azimuthally in the direction of increasing θ) and $\hat{z}$ (pointing in positive z).

The line element vector is,

$$d\vec{\ell} = dr \hat{r} + r d\theta \hat{\theta} + dz \hat{z} \equiv (dr, r d\theta, dz).$$

The surface element on a surface of constant z is $dS = r dr d\theta$ and on a surface of constant radius is $dS = r d\theta dz$. The volume element is $dV = r dr d\theta dz$.

Relating Cartesian and polar coordinates

How can we relate coordinates and unit vectors in the polar system to the Cartesian system? In the z -direction, this is trivial since they are the same in both systems. By trigonometry in the xy -plane, we relate Cartesian and polar coordinates through the relations,

$$x = r \cos \theta, \quad y = r \sin \theta.$$

These relations in turn lead to the useful results,

$$r^2 = x^2 + y^2, \quad \tan \theta = y/x$$

Similarly, we can express the polar unit vectors $\hat{r}$ and $\hat{\theta}$ in terms of the Cartesian unit vectors as,

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y} \equiv (\cos \theta, \sin \theta, 0), \quad \hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y} \equiv (-\sin \theta, \cos \theta, 0).$$

Note that the fact that the polar unit vectors, $\hat{r}$ and $\hat{\theta}$, depend on one of the coordinates, θ , is important. It means that we have to take care when integrating/differentiating vectors in cylindrical polar coordinates, since these unit vectors cannot be taken as constant during the integration/differentiation process. You should go to Appendix A.1 to familiarise yourself with this issue and how to deal with it.

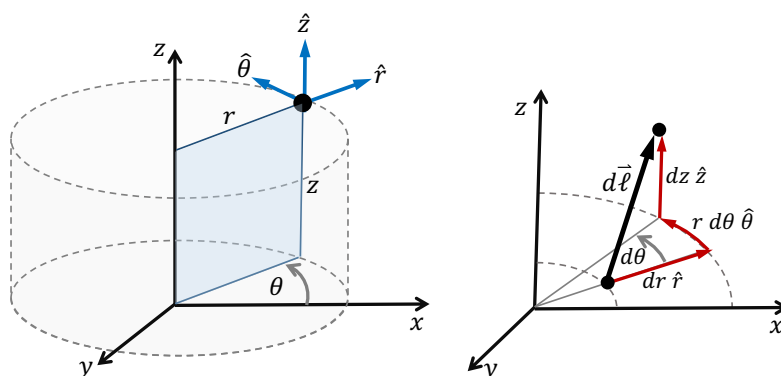
2.6 Density

Density is the mass per unit volume of a material. It tells us how much mass there is in a given volume.

The mass dM of an infinitesimal volume dV is $dM = \rho dV$, where ρ is the **density**.

For some larger volume V , in which the density varies with position $\rho = \rho(\vec{x})$, the mass M is found by integrating over all of the infinitesimal volumes,

$$M = \iiint_V \rho(\vec{x}) dV.$$



Left: Cylindrical polar coordinates, with the unit vectors shown (blue arrows). Right: The line element vector $d\vec{\ell}$.

Meanwhile, if the density is constant within V we simply have $M = \rho V$.



EXAMPLE 3: A material has a density which varies in space as $\rho(\vec{x}) = y^2 + z^2$.

- Evaluate the mass of the cubic volume $0 \leq x, y, z \leq 1$.
- Evaluate the mass of the cylindrical volume $0 \leq r, z \leq 1$. Use cylindrical polar coordinates and the definite integral $\int \sin^2 \theta \, d\theta = \frac{1}{2} \left(\theta - \frac{1}{2} \sin 2\theta \right) + C$.

2.7 The continuum approximation

We will describe the fluid (gas or liquid) in terms of its density ρ and its flow velocity $\vec{v}$, both of which can vary in time and space,

$$\rho = \rho(\vec{x}, t), \quad \vec{v} = \vec{v}(\vec{x}, t).$$

Together they describe how much stuff there is at a given point and time, and how it moves.

The **continuum assumption** treats the fluid as a continuous distribution^a of matter whose properties are defined at any point in space and time, and vary continuously.

^aSometimes called a *continuum*.

Properties include the velocity, density, pressure or temperature of the fluid.

The continuum approximation is the fundamental assumption behind fluid dynamics, and allows us to apply calculus (e.g. to take derivatives). However, it ignores the true fine-scale, “microscopic” structure of the fluid. Take the air in the room as an example. On the microscopic scale, the gas is composed of molecules and atoms (mostly nitrogen and oxygen molecules), all flying around at random, and with empty space in between. On this scale the fluid and its properties are certainly not continuous.

As humans, we don’t perceive this fine scale structure - we sense the temperature and velocity of the gas (e.g. a draught in the room) as smoothly and continuously varying. This is because our senses measure the properties over larger, “macroscopic” scales where we sense the average behaviour of many particles (rather than individual particles themselves). Such averaging is central to the continuum approximation.

How big must the averaging region be for the continuum approximation to be valid? A

cube of air which is 1 micrometer[‡] wide contains around 10 million particles. The same volume of a liquid, which is denser, will contain even more particles. Thus for the majority of fluid dynamics situations we might be interested in there are more than enough particles to ensure good averaging, and so the continuum approximation is valid[§].

A **fluid parcel**^a is a volume of fluid which is sufficiently large to satisfy the continuum approximation but small enough to treat as infinitesimal with respect to the fluid properties of interest.

^aSometimes *fluid particle*.

Why don't we just model the atoms and molecules themselves, treating each as an object moving according to Newton's second law of motion and the forces they experience (e.g. from collisions with other particles or the container walls)? We would need to solve at least one ordinary differential equation per particle, and so a 1-micrometer-cube of air would require solving many millions of ODEs, a thoroughly intractable task! The futility of modelling any sizeable lump of matter on a particle-by-particle basis is the reason we adopt the continuum approximation, and why it is so important.

2.8 The microscopic reality of macroscopic properties

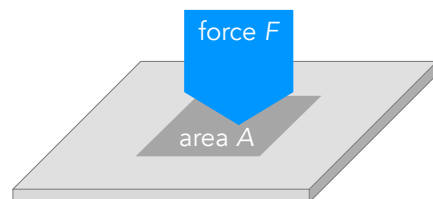
The macroscopic properties of a fluid we observe and measure are caused by the (average of the) microscopic behaviour of the particles. Here are some examples.

Flow. Picture a volume of fluid flowing in a certain direction, such as down a stream or river. In fact, the particles within this volume are moving with different speeds and directions, some even moving backwards. Averaging over all of these individual movements give the overall motion of the fluid volume.

Pressure. Recall the school definition of pressure. Consider a force F being exerted normal to some surface of area A ; the pressure is then defined as

$$P = F/A. \quad (2.1)$$

Usually this is introduced as a mechanical force on a surface, such as by a hammer or thumb, but fluids can exert pressure too.



The school definition of pressure.

A fluid in some container will exert a pressure on the container surfaces. The more fluid we cram into the volume, the greater the pressure. Microscopically, the pressure results from the collision of many particles off the surface. As a particle in the fluid collides and bounces off the surface, it exerts a tiny pressure on the surface; average this over all the particles in the fluid and one experiences the overall pressure. If we increase the temperature of the fluid, the pressure increases because the particles collide with greater frequency and because each collision exerts a greater force.

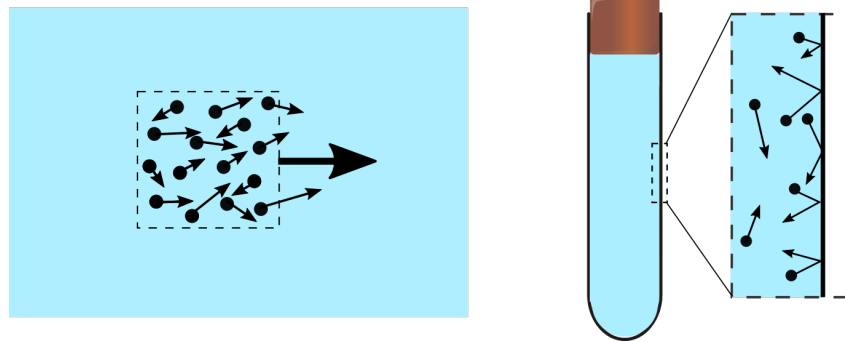
It also follows from this that the fluid pressure is zero in a vacuum.

Temperature. Temperature is related to how vigorously the particles move, on average. At any given temperature, there is a distribution across speeds - most particles move at

[‡]1 micrometer= 10^{-6} m, about $\frac{1}{10}$ th the width of a biological cell.

[§]One area where the continuum approximation must be considered carefully is for high-altitude planes and spacecraft, where the ambient air is very dilute.

close to the average speed, but a small fraction can have a much higher speed. These energetic ones are the ones associated with evaporation. Increasing the temperature increases the average kinetic energy. Our sensation of temperature is related to sensing the bombardment of particles on our skin. If these particles are too vigorous, they burn us.



Left: a volume of fluid moving to the right is actually composed of particles which move in many directions and speeds. Right: the pressure on the walls of the tube due to the gas within is due to the repeated bombardment of the particles on the wall.

3 Flows

We now develop our description of fluids based on the continuum approximation. In this chapter we focus on describing how the fluid *flows* (we ignore the fluid density ρ).

3.1 Flow velocity

We describe the flow by $\vec{v} = \vec{v}(\vec{x}, t)$, the **flow velocity** at position $\vec{x}$ and time t .

In Cartesian coordinates $\vec{v}$ has the general form,

$$\vec{v} = (v_x(x, y, z, t), v_y(x, y, z, t), v_z(x, y, z, t)),$$

where v_x , v_y and v_z are the components of $\vec{v}$ in each direction. We can visualise $\vec{v}$ through a quiver plot, in which arrows depict $\vec{v}$ at regularly-spaced points.

3.2 1D and 2D flows

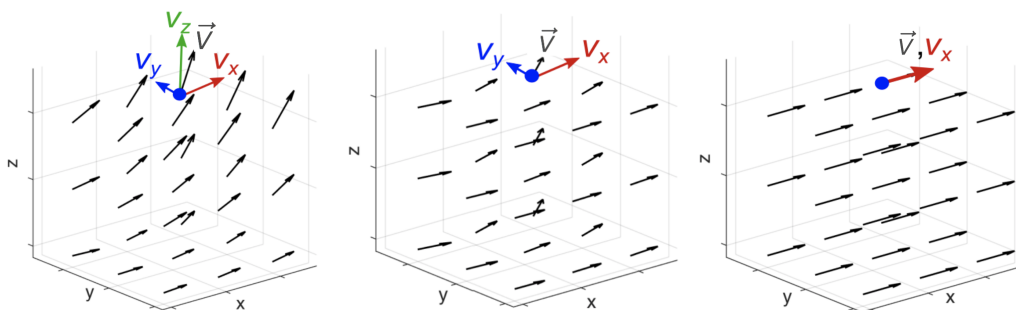
In general a flow is three-dimensional, varying with all three directions. However, in certain scenarios the flow can be approximated as two-dimensional (2D) or one-dimensional (1D), which allows for a simpler mathematical treatment.

A **2D flow** is a function of two spatial coordinates only, and has zero component in the remaining direction.

For example, $\vec{v} = (v_x(x, y, t), v_y(x, y, t), 0)$ is a 2D flow, which we may choose to write more simply as $\vec{v} = (v_x, v_y)$.

A **1D flow** is a function of one spatial coordinate, and has no component in the remaining direction.

For example, $\vec{v} = (v_x(x, t), 0, 0)$ is a 1D flow, or more simply denoted as $\vec{v} = v_x$.



3D flow (left), 2D flow (middle) and 1D flow (right).

3.3 Flows in cylindrical polar coordinates

We will sometimes work with flows in cylindrical polar coordinates (see Section 2.5). Then $\vec{v}$ has the general form,

$$\vec{v} = (v_r(r, \theta, z, t), v_\theta(r, \theta, z, t), v_z(r, \theta, z, t)),$$

where v_r is the radial velocity component (along $\hat{r}$), v_θ is the azimuthal component (along $\hat{\theta}$) and v_z is the vertical component (along $\hat{z}$).

Flows in cylindrical polar coordinates can also be 1D, 2D or 3D, and we will most often encounter the case of 2D flow in (r, θ) .

3.4 Uniform flows

In a **uniform flow**, $\vec{v}$ is the same at every point in the flow.

It follows that for a uniform flow $\vec{v}$ satisfies,

$$\frac{\partial \vec{v}}{\partial x} = \vec{0}, \quad \frac{\partial \vec{v}}{\partial y} = \vec{0}, \quad \frac{\partial \vec{v}}{\partial z} = \vec{0}.$$

3.5 Steady flows

In a **steady flow** the flow velocity does not change with time; in an **unsteady flow** it does.

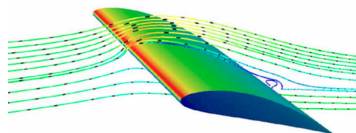
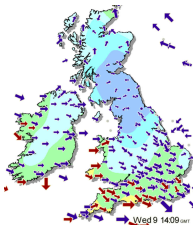
A steady flow implies that $\frac{\partial \vec{v}}{\partial t} = \vec{0}$. However, $\vec{v}$ may change with position $\vec{x}$.

In practice, for some steady flow, there is always slight variations in velocity and pressure, etc, but if the average values are constant, then flow is considered steady.



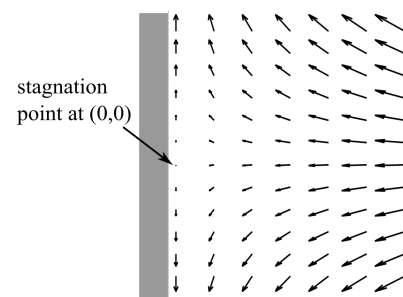
EXAMPLE 4: For the following flows, state whether they are steady or unsteady, uniform or non-uniform, and the flow dimension. For the images, give an approximate classification.

- (a) $\vec{v} = (x, -y, 0)$ (b) $(xy, t, x + y)$ (c) $(1, 2, 3)$ (d) $(0, t \cos \theta, z)$
 (e) (f) (g) (h)



3.6 Stagnation points

A **stagnation point** is a point $\vec{x}$ in the fluid where $\vec{v} = \vec{0}$.



At a stagnation point, we say the flow *stagnates*. A fluid parcel cannot escape a stagnation point; this is because the fluid there has no velocity component to carry it away.

Consider, as an example, the 2D flow $\vec{v} = (v_x, v_y) = (-x, y)$, shown here as a quiver plot. This represents a flow from right to left and incident upon a hard wall. The flow clearly stagnates at $(0, 0)$.



EXAMPLE 5: For the following 2D steady flows, identify the location of any stagnation points.

- (a) $\vec{v} = (v_x, v_y) = (x^2, y - 1)$
- (b) $\vec{v} = (v_x, v_y) = (x - y^2, xy)$
- (c) $\vec{v} = (v_x, v_y) = (2, x)$
- (d) $\vec{v} = (v_r, v_\theta) = (r - 1, \cos \theta)$.

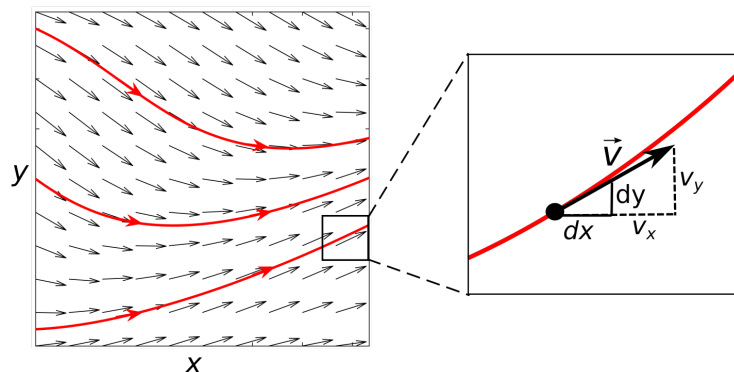
3.7 Streamlines

Streamlines are an important way to visualise the flow.

A **streamline** is a curve which is everywhere tangential to the instantaneous flow velocity.

A streamline gives an instantaneous snapshot of the flow. They have a direction associated with them, which we depict this by adding arrows to the curves.

For steady flow, streamlines are fixed over time, while for unsteady flow the streamlines will change with time. Along a streamline the fluid is moving in the same direction as the line, and so the fluid cannot cross it. This also means that two streamlines can not cross.



Some streamlines in a 2D flow and a zoom-in on a region of the flow.

Streamlines are parametric curves $\vec{x}(s) = (x(s), y(s), z(s))$, with parameter s , which satisfy,

$$\frac{d\vec{x}}{ds} = \vec{v}(\vec{x}(s), t). \quad (3.1)$$

Equation (3.1) embodies that the curve $\vec{x}(s)$ should have the same direction as $\vec{v}$ at

each point in space, all at some fixed time t . It has three components,

$$\frac{dx}{ds} = v_x, \quad \frac{dy}{ds} = v_y, \quad \frac{dz}{ds} = v_z. \quad (3.2)$$

Solving these equations gives the general solution for $\vec{x}(s)$, which describes a family of streamline curves. The direction of the streamline is the direction of increasing s . We can find a particular streamline by imposing a condition, such as that $\vec{x}(s=0) = (x_0, y_0, z_0)$. Note that if a velocity component is zero, then that coordinate stays constant along the streamline, e.g. for a 2D flow with $v_z = 0$, $z(s)$ is a constant.

For a 2D flow $\vec{v} = (v_x, v_y)$ the streamline curves $y(x)$ satisfy,

$$\frac{dy}{dx} = \frac{v_y}{v_x}. \quad (3.3)$$

This relation can be obtained by dividing the second equation in Eq. (3.2) by the first one. The relation can also be understood geometrically as shown in the figure; the gradient of the streamline corresponds to v_y/v_x .

Note that, at a stagnation point, streamlines are undefined since the flow velocity vanishes there.

Equation (3.3) provides an alternative to the parametric method when seeking the streamlines for a 2D flow. However, it tells us nothing about the direction of the streamlines; we must infer this by considering the direction of $\vec{v}$.



EXAMPLE 6: Consider the 2D flow $\vec{v} = (\alpha x, -\alpha y)$ with $\alpha > 0$.

- Use Eq. (3.3) to determine the equation $y(x)$ for the streamlines.
- Sketch some typical streamlines in the xy plane, including arrows to show the flow direction (you will need to deduce the flow direction from considering $\vec{v}$). Highlight the streamline which goes through $(x_0, y_0) = (-1, -1)$.
- Determine the parametric representation of the streamlines $\vec{x}(s) = (x(s), y(s))$, and confirm that it gives the same results as part (a).

3.8 Pathlines

A **pathline** is the trajectory $\vec{x}(t)$ of a fluid parcel in time as it moves with the flow and is defined by,

$$\frac{d\vec{x}(t)}{dt} = \vec{v}(\vec{x}(t), t), \quad (3.4)$$

The interpretation of this definition is that, at each point in time, the instantaneous velocity of the fluid parcel must equal the flow velocity at that point and time. Writing out the three components we have,

$$\frac{dx}{dt} = v_x, \quad \frac{dy}{dt} = v_y, \quad \frac{dz}{dt} = v_z. \quad (3.5)$$

Solving these equations gives the general solution of the pathlines. A particular pathline can be identified by imposing an initial condition $\vec{x}(t=0) = (x_0, y_0, z_0)$.

For steady flows, streamlines always coincide with pathlines. If the flow is unsteady, streamlines and pathline may differ, and they may be very different. Whereas streamlines represent an instantaneous snapshot of the flow, pathlines instead have a history associated with them.



EXAMPLE 7: Consider the 2D unsteady flow $\vec{v} = (\alpha x t, -\alpha y)$ for $\alpha > 0$.

- Find $y(x(t))$ for the streamlines (keeping your answer as a function of t).
- Find the equation $\vec{x}(t)$ of the pathlines, with general initial conditions (x_0, y_0) .
- Find the streamline and the pathline which go through the point $(1, 2)$ at $t = 1$.

3.9 Eulerian and Lagrangian views

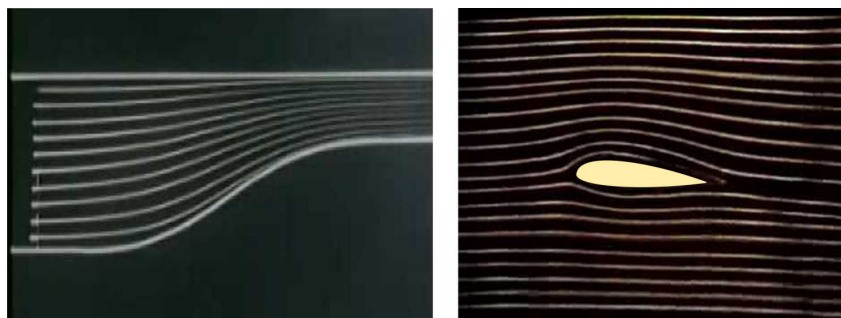
There are two ways in which we can describe the flow, depending on our perspective.

- In **Eulerian view**, the fluid is viewed at a fixed point/s in space. An analog would be the traffic analyst who stands at a road and monitors the flow of traffic past that point.
- In the **Lagrangian view**, the flow is viewed at a fluid parcel which moves through the fluid. The analog would be the traffic analyst who monitors traffic while driving with the traffic.

Streamlines correspond to the Eulerian perspective, since we are representing the flow in a fixed frame of reference (the coordinate system). Meanwhile, pathlines correspond to the Lagrangian description since the fluid is being viewed from the perspective of a parcel moving with the fluid.

3.10 Material derivative

It is useful to know how a property of the flow, which could be the density, temperature or the concentration of some chemical, changes from the perspective of a fluid parcel moving in the flow. One might think of this as the rate of change from the Lagrangian



Experimental images of fluid flow (from left to right) in which light-coloured dye is introduced at various vertical points on the left. The dye is carried with the flow to show some pathlines. The flow is steady and so the pathlines also correspond to streamlines. On the left, the flow is in a channel which passes into a constriction; the corresponding movie can be found at <https://www.youtube.com/watch?v=DOUfyDHxkYQ>. On the right, the flow passes an aerofoil.

perspective. This is provided by the *material derivative*^{*}.

The **material derivative** of a scalar function $f = f(\vec{x}, t)$ tells us the rate of change of f as experienced by a fluid parcel moving with the flow^a, and is given by,

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + (\vec{v} \cdot \nabla)f. \quad (3.6)$$

^aThat is, along a pathline.

If $Df/Dt = 0$ then f is constant for a particular fluid parcel; however, different fluid parcels may have different values of f .

The two parts of the material derivative have different meanings:

- $\partial f/\partial t$ is the rate of change of f at a fixed point in space. This is the Eulerian rate of change of f .
- $(\vec{v} \cdot \nabla)f$ is the change in f as a result of the fluid parcel moving to a new location where f is different (often termed the “convective” or “advective” part).

For steady flow ($\partial f/\partial t = 0$), the rate of change of f following a fluid parcel (which now travels along a streamline) is $(\vec{v} \cdot \nabla)f$. From this we can make the following statement.

For steady flow, and if $(\vec{v} \cdot \nabla)f = 0$, then f is constant along a streamline.

Derivation of the material derivative

The fluid parcel moves along a pathline as per Eq. (3.4) with position $\vec{x}(t)$. Let’s compute df/dt as experienced by this fluid parcel. Since x , y and z depend on t we employ the chain rule,

$$\frac{d}{dt}f[x(t), y(t), z(t), t] = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}. \quad (3.7)$$

$x(t)$, $y(t)$ and $z(t)$ evolve according to Eq. (3.5) and so we can write,

$$\frac{d}{dt}f[x(t), y(t), z(t), t] = \frac{\partial f}{\partial t} + v_x \frac{\partial f}{\partial x} + v_y \frac{\partial f}{\partial y} + v_z \frac{\partial f}{\partial z}, \quad (3.8)$$

which is equivalent to,

$$\frac{d}{dt}f[x(t), y(t), z(t), t] = \frac{\partial f}{\partial t} + (\vec{v} \cdot \nabla)f. \quad (3.9)$$

where the right side is the material derivative.

Material derivative in cylindrical polar coordinates

Since in cylindrical polar coordinates the gradient operator is defined as $\nabla = \left(\frac{\partial}{\partial r}, \frac{1}{r} \frac{\partial}{\partial \theta}, \frac{\partial}{\partial z} \right)$ we can write the material derivative of a scalar function $f = f(\vec{x}, t)$ as,

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + v_r \frac{\partial f}{\partial r} + \frac{v_\theta}{r} \frac{\partial f}{\partial \theta} + v_z \frac{\partial f}{\partial z}. \quad (3.10)$$

^{*}There is an unfortunately long list of other terms for this derivative: total derivative, convective derivative, substantial derivative, hydrodynamic derivative, Lagrangian derivative, particle derivative.... Why not make up your own?



EXAMPLE 8: A 2D flow moves as $\vec{v} = (\alpha x, -\alpha y)$, with $\alpha > 0$. The concentration of a chemical in the flow behaves in time and space as $c(x, y, t) = \beta x^2 y e^{-\alpha t}$. Does the concentration change for a particular fluid parcel over time?

3.11 Acceleration of a fluid parcel

The **fluid acceleration** is the material derivative of the flow velocity,

$$\vec{a} = \frac{D\vec{v}}{Dt}. \quad (3.11)$$

The fluid acceleration is the acceleration experienced by a fluid parcel moving with the flow. It is the material derivative of the fluid velocity.

Note that the material derivative Eq. (3.9) is defined to act on a scalar function. When applying it to the velocity, a vector, we should consider applying the material derivative to each component of the velocity, which are scalar functions. Performing this in Cartesian coordinates gives,

$$\frac{D}{Dt} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \frac{\partial}{\partial t} \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} + (\vec{v} \cdot \nabla) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \begin{pmatrix} \frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \end{pmatrix}$$

$\frac{D\vec{v}}{Dt}$ gives the total acceleration vector experienced by whatever parcel happens to be at the location and time of interest. $\frac{\partial \vec{v}}{\partial t}$ is the local acceleration; it is zero for a steady flow. The $(\vec{v} \cdot \nabla)\vec{v}$ term is the acceleration due to the movement of a fluid parcel to a different part of the flow field; it is zero for a uniform flow.

Thus for a flow which is both steady and uniform it follows that $\vec{a} = \vec{0}$. However, a flow which is steady $\vec{a}$ can be non-zero, since the fluid parcel may move to a position where $\vec{v}$ has a different value (due to the convective term). Take for example $\vec{v} = (v_x, v_y) = (3x, 3y)$, for which $\vec{a} = (9x, 9y) \neq 0$.

3.11.1 Fluid acceleration in cylindrical polar coordinates

The fluid acceleration in cylindrical polar coordinates, applied to a velocity field $\vec{v} = (v_r, v_\theta, v_z)$, is

$$\begin{aligned} \frac{D\vec{v}}{Dt} = & \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z \frac{\partial v_r}{\partial z} \right) \hat{r} \\ & + \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta v_r}{r} + v_z \frac{\partial v_\theta}{\partial z} \right) \hat{\theta} \\ & + \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) \hat{z} \end{aligned}$$

The derivation of this result is more complicated than the Cartesian result. This is because one must account for the fact that the radial unit vector $\hat{r}$ and azimuthal unit vector $\hat{\theta}$ depend on azimuthal angle θ . The full derivation can be found in Appendix A.2.



EXAMPLE 9: Determine the acceleration of a fluid parcel in the flows:

- (a) $\vec{v} = (v_x, v_y, v_z) = (-3x, -3y, 6z)$
- (b) A uniformly rotating fluid with fixed angular velocity Ω , $\vec{v} = (v_r, v_\theta, v_z) = (0, \Omega r, 0)$

3.12 Vortices and vorticity

Vortices

A basic component of fluid motion is the swirling of the fluid.

A **vortex** is a rotating flow of the fluid about some axis.

Examples are tornados, plughole vortices and the wingtip vortices of aeroplanes. Flow past an obstacle can create an eddy in the fluid, such as when a river flows past a rock. On larger scales, the oceans swirl due to currents. Here we will introduce vortical flows and ways to quantify them.



In many real-world situations, fluids undergo swirling, rotational motions. These are vortices.

Vorticity

The **vorticity** of the flow is $\vec{\omega} = \nabla \times \vec{v}$.

The vorticity gives a *local* measure of the fluid's rotation. Picture a small cross floating in the flow. If the arms of the cross rotate, there is vorticity at that point in space. The faster it rotates, the larger the vorticity there.

A flow is **irrotational** if $\vec{\omega} = \vec{0}$. Otherwise, it is **rotational**.

We will study irrotational flows in detail in Chapter 6.

Expanding it out into its components, the vorticity of $\vec{v} = (v_x, v_y, v_z)$ is,

$$\vec{\omega} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}, \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}, \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right). \quad (3.12)$$

For a 2D fluid flow $\vec{v} = (v_x, v_y, 0)$, $\vec{\omega}$ reduces to,

$$\vec{\omega} = \left(0, 0, \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right). \quad (3.13)$$

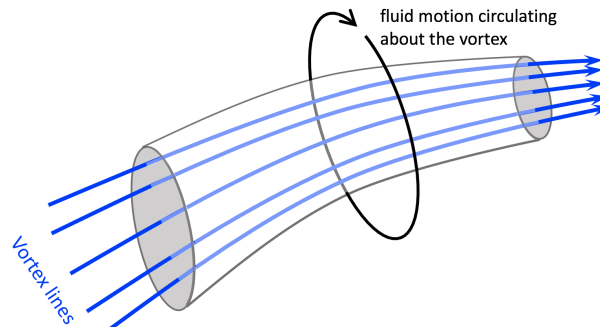
Note how $\vec{\omega}$ points perpendicular to the flow plane.

Vortex lines

A **vortex line** is a curve which is tangent to the vorticity vector.

The vortex line shows us where the flow is rotating within the fluid; note that the flow rotates perpendicular to the vortex line. For a tornado, for instance, the vortex lines will be clustered along the core of the tornado, pointing along its axis.

A vortex line is analogous to a streamline, since a streamline is tangent to the velocity vector.



Vortex lines along a vortex.



EXAMPLE 10: Calculate the vorticity $\vec{\omega}$ of the following flows:

- (a) $\vec{v} = (v_x, v_y, v_z) = (\beta y, 0, 0)$, with β a constant.
- (b) $\vec{v} = (v_r, v_\theta, v_z) = (0, \Omega r, 0)$, with Ω a constant.

3.13 Circulation

The **circulation** Γ about the closed curve C , with line element vector $d\vec{\ell}$, is,

$$\Gamma = \oint_C \vec{v} \cdot d\vec{\ell}. \quad (3.14)$$

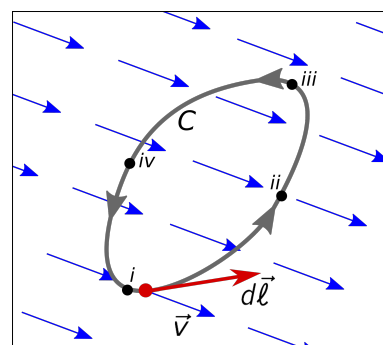
The circulation is the integral of $\vec{v} \cdot d\vec{\ell}$, the component of the fluid velocity which is tangential to the curve. Where $\vec{v}$ is parallel to the curve (point *i*), the contribution of $\vec{v} \cdot d\vec{\ell}$ is positive. Where $\vec{v}$ is anti-parallel to the curve (point *ii*), the contribution is negative. Where $\vec{v}$ is perpendicular to the curve (points *iii* and *iv*), the contribution of $\vec{v} \cdot d\vec{\ell}$ is zero.

If C is some rectangular or square path, we can break the line integral up into 4 contributions, one for each side, and proceed to evaluate each.

If C is a circle of radius R , at constant z , we can work in cylindrical polar coordinates with the line element vector $d\vec{\ell} = (dr, r d\theta, dz)$ set as,

$$d\vec{\ell} = (0, R d\theta, 0),$$

and integrating over $0 \leq \theta \leq 2\pi$.



Circulation about a path C .



EXAMPLE 11: Evaluate the circulation of:

- (a) $\vec{v} = (v_x, v_y) = (x^2 - y^2, 2xy)$, anti-clockwise about a rectangular path defined by the four points $(0, 0)$, $(a, 0)$, (a, b) and $(0, b)$.
- (b) $\vec{v} = (v_r, v_\theta) = (0, \Omega r)$, anti-clockwise about a circular path of radius R .

3.14 Relating vorticity and circulation

Whereas the vorticity is a vector field that gives a local measure of the rotation, the circulation is a scalar quantity that gives a measure of the rotation over a finite region of the fluid. Nonetheless, circulation and vorticity are related.

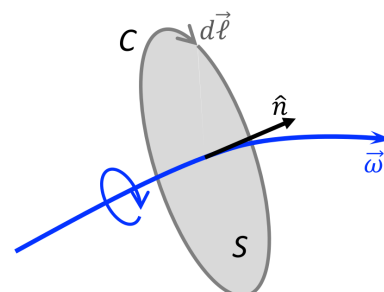
The circulation around a curve C is equal to the flux^a of $\vec{\omega}$ through the surface S bounded by C .

^a“Flux” is the integral of a vector field normal to a surface.

We can show this using Stokes’ theorem - see Appendix A.4 - which allows us to write a line integral in terms of a surface integral. Using Stokes’ theorem we can write the circulation as,

$$\Gamma = \oint_C \vec{v} \cdot d\vec{\ell} = \iint_S \vec{\omega} \cdot \hat{n} \, dS.$$

The surface integral integrates the component of the vorticity normal to the surface, i.e. the flux of $\vec{\omega}$. It follows that the vorticity at a point is the circulation per unit area.



Vorticity (denoted by the blue vortex line) passes through the surface S bounded by C .



EXAMPLE 12: For a flow $\vec{v} = (v_x, v_y, v_z) = (ay, 0, 0)$, where a is a constant, compute the circulation anti-clockwise about the square path defined by the vertices $(0, 0)$, $(L, 0)$, (L, L) and $(0, L)$. Check the result by using Stokes’ theorem.

3.15 Example flow: solid-body rotation

In **solid-body rotation** the flow is azimuthal with speed $v = \Omega r$, where Ω is a constant.

In this flow, each part of the fluid moves (in a circle) with the same angular speed, given by Ω . The angular speed is the rate of change of angle, $\Omega = \frac{d\theta}{dt}$. Examples of this flow type are the steady flow of water in a rotating bucket and a rotating disc.

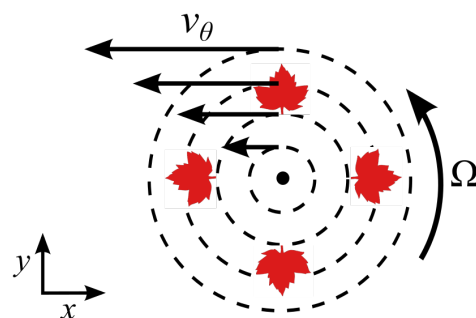
This flow is most conveniently expressed in cylindrical polar coordinates as,

$$\vec{v} = (v_r, v_\theta, v_z) = (0, \Omega r, 0).$$

In Cartesian coordinates the same flow is $\vec{v} = (v_x, v_y, v_z) = (-\Omega y, \Omega x, 0)$.

Since Ω is constant, each fluid parcel takes the same time to sweep out a full orbit of 2π ; this gives the period $T = 2\pi/\Omega$. Note how the azimuthal speed v_θ must increase with radius to ensure this.

The circulation for solid-body rotation about a circle of radius R is $\Gamma = 2\pi\Omega R^2$. The vorticity is $\vec{\omega} = (0, 0, 2\Omega)$; note how it is uniform in space. Picture of a leaf (representing a fluid parcel) moving with the flow: as the leaf orbits the origin, its outer part travel with higher speed than the inner part, such that the leaf rotates once every orbit (irrespective of where it is positioned radially), hence $\vec{\omega}$ is uniform.



Solid-body rotation.

3.16 Example flow: shear flow

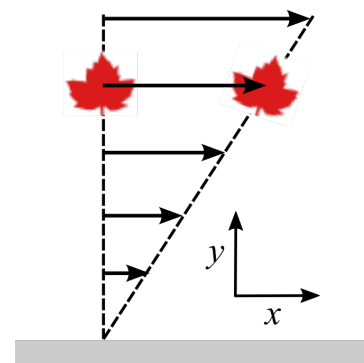
In a **shear flow** adjacent layers of fluid move parallel to each other at different speeds.

This type of motion happens when a fluid flows past a fixed surface: at the surface $\vec{v}$ must be zero, but away from the surface the flow will reach some ambient value. It can also arise when two regions of fluid with relative speeds meet.

An example is the 2D flow,

$$\vec{v} = (v_x, v_y, v_z) = (\beta y, 0, 0),$$

with constant β . The flow is along x but its speed changes linearly with y . Surprisingly, the vorticity $\vec{\omega} = (0, 0, -\beta)$ is not zero, even though the particles are flowing along straight, parallel paths. This can be understood by considering the leaf: as it is carried downstream it will rotate since its top part is being carried at a faster speed than its bottom part.



Shear flow.

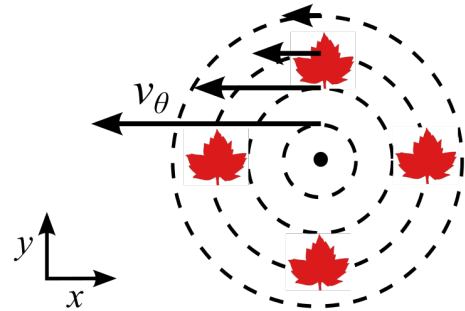
3.17 Example flow: free vortex

For a **free vortex** the flow is azimuthal with speed a/r , where a is a constant.

We can write the flow in polar coordinates as,

$$\vec{v} = \left(0, \frac{a}{r}, 0\right),$$

where a is a constant. The velocity is not defined at $r = 0$. Weirdly, the vorticity is zero, despite the fact that the fluid flows around the axis. How can this be? A leaf orbits in the flow, but its outer part travels slower than the inner part such that the leaf keeps the same orientation as it orbits (a bit like the gondolas on a Ferris wheel), hence $\vec{\omega} = \vec{0}$.



Free vortex.

3.18 Example flow: tornados

For a real tornado (assumed to be aligned along z), the wind's velocity is purely azimuthal around the axis, of the form $\vec{v} = (0, v_\theta(r), 0)$. The vorticity has the general form $\vec{\omega} = (0, 0, \omega_z(r))$. Within the vortex "core", the flow is like solid-body rotation: $v_\theta \propto r$ and $\omega_z = \text{constant}$. Away from the core, the flow is like the free vortex: $v_\theta \propto 1/r$ and $\omega_z = \text{constant}$. In between these inner and outer regions, the behaviour smoothly crosses between them - see Figure 3.1. Note how the vorticity is concentrated within the core of the tornado.

3.19 Example flow: the Rankine vortex

A mathematical model for a tornado is the Rankine vortex,

$$v_\theta(r) = \begin{cases} \Omega r & \text{when } r < a \\ \frac{\Omega a^2}{r} & \text{when } r > a. \end{cases}$$

Unlike a real tornado, the vorticity and azimuthal flow suddenly change between the inner and outer behaviour, but nonetheless the Rankine vortex is a useful approximation to real tornados.

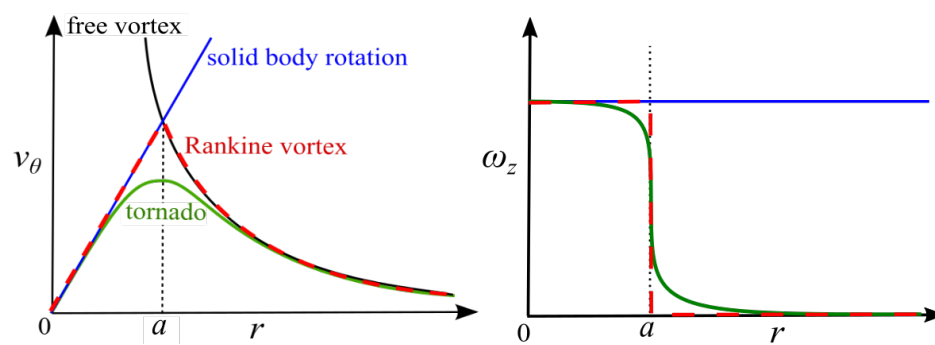


Figure 3.1: Comparison of how the azimuthal velocity component and axial vorticity component vary with radius for various types of rotating flows.

4 Static fluids

The study of fluids which are not moving - sometimes called “hydrostatics” - can reveal many important phenomena, such as why things float and how hydraulic jacks work. Here we will introduce this topic, including what we mean by fluid pressure, what forces can exist in a fluid, and how buoyancy arises.

We will consider the general case where the fluid density may vary in space, i.e. $\rho = \rho(\vec{x})$. We will also consider the simpler case where the fluid density is constant across the whole fluid; such fluids are known as **incompressible fluids**. Liquids, e.g. water and oil, can be treated as incompressible fluids under most typical conditions.

4.1 Forces within the fluid

The forces within the fluid determine how it moves, or whether it stays static. There are three main types of forces which may act on a fluid parcel:

Pressure force: the force due to the pressure exerted by the neighbouring fluid parcels.

Body forces: forces which act on the fluid parcel from an external force field, such as gravity.

Viscous forces: frictional forces exerted on the parcel by neighbouring parcels which are moving relative to it. We neglect viscosity throughout this module, and so this type of force is not relevant to us.



ASIDE: *Viscosity* is related to how “thick” a fluid is: honey is more viscous than water. But even water experiences significant viscosity - if we stir a glass of water and then stop, the swirling flow slows down quickly due to viscosity.

Including viscosity in our mathematical description of fluids is possible, but makes the equations difficult to solve. Although we ignore viscosity in this module, we can still describe many important behaviours of fluids in the real world.

Fluids without viscosity do exist. “Superfluids” can be formed in laboratories when certain fluids are cooled down towards absolute zero. The absence of viscosity means that if we stop stirring them they will continue to flow forever!

4.2 Pressure and the pressure force

4.2.1 Pressure

A surface within the fluid (either a real surface from an object or wall, or some virtual surface) will experience forces from the neighbouring fluid; these forces lead to *pressure*. It is worth remembering from Section 2.8 that, microscopically, these forces are due to the bombardment of the surface by the particles which make up the fluid.

The **infinitesimal pressure force** $d\vec{F}$ acting on a surface element dS^a within the fluid is,

$$d\vec{F} = -P \hat{n} dS, \quad (4.1)$$

where P is the pressure of the fluid at that point and $\hat{n}$ is the unit outward vector normal to dS .

^aRecall that the surface element has area dS .

We see from this definition that pressure P is the proportionality constant which relates the force on a surface to its area. The negative sign appears since the force acts inwards onto the surface, whereas $\hat{n}$ points outwards by definition.

Note that in the definition (4.1) the surface elements are infinitesimally small such that we take i) them to be flat, and ii) the pressure to be constant. Thus, if the whole of S is flat and experiences constant pressure, we can take dS to be the whole surface of interest, giving,

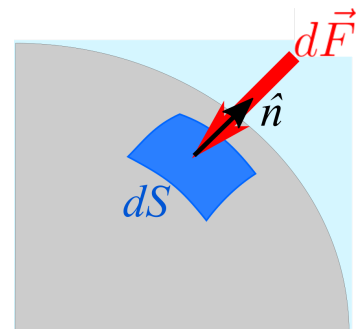
$$\vec{F} = -P \hat{n} A,$$

where A is the area of the surface. Taking the magnitude we obtain,

$$F = PA,$$

which recovers the school definition of pressure we introduced in Equation (2.1).

Pressure is a scalar quantity (it has no direction, although it gives rise to forces which do have directions). In general, it may vary throughout the fluid and in time, $P = P(\vec{x}, t)^*$, and it is never negative.



Pressure force $d\vec{F}$ on a surface element dS .

Pressure varies continuously throughout a fluid or connected fluids.

The continuity of pressure is an important property, particularly across fluids in contact. For example, the pressure on the surface of a glass of water, which is open to the atmosphere, will equal that of the atmospheric air bordering the water's surface due to the continuity of pressure. (If however, an air-tight lid is placed on the glass such that the water and air are no longer in contact, then this continuity of pressure is broken.)

4.2.2 Pressure Force

The **pressure force** exerted on a surface S by the surrounding fluid with pressure $P = P(\vec{x})$ is,

$$\vec{F} = - \iint_S P \hat{n} dS. \quad (4.2)$$

*In this module we will only consider pressure to vary in space, not time

This follows from Eq. (4.1): we are summing the force contributions $d\vec{F}$ from all the elements dS that comprise S . Because the pressure force depends on the surface area involved, it is often classed as a *surface force*.

We use Eq. (4.2) to calculate the pressure force acting on *any* surface in or contacting the fluid. These may be open surfaces or closed surfaces, virtual surfaces within the fluid or real surfaces (such as the surface of an object or a container).



EXAMPLE 13: Consider a cubic volume of fluid, with sides of length $L = 1$, within an ambient fluid. The bottom of the cube lies on the $z = 3$ plane. The pressure varies throughout the fluid as $P(\vec{x}) = z^2$. Find the pressure force on (a) the bottom surface of the cube, (b) the top surface of the cube, (c) one (any) side of the cube, and (d) the net force on the cube.

4.2.3 Pressure Force on a Volume

The **pressure force on a fluid volume** V , enclosed by the closed surface S and lying within the fluid, can be written as,

$$\vec{F} = - \iiint_V \nabla P \, dV. \quad (4.3)$$

From this we can say that for an infinitesimally-small fluid parcel of volume dV the pressure force imposed by the surrounding fluid follows from Eq. (4.3) as,

$$d\vec{F} = -\nabla P \, dV. \quad (4.4)$$

This tells us that a gradient in pressure implies a force directed towards lower pressure. A real-life example of this effect is when the valve on an inflated tyre is opened. Air accelerates out of the valve, first quickly (due to the large pressure gradient) and then progressively slower (as the pressure difference reduces). Once the pressure is equalized the flow ceases.

Note how the result of Eq. (4.3) can only be applied to volumes *lying within the fluid*, that is, away from any boundaries. This is a requirements of its derivation (see below). It means that if we want to calculate the pressure force in some region which does involves boundaries then we should usually resort to the calculating the pressure force as a surface integral via Eq. (4.2).

Derivation

Recall Gauss' theorem for a scalar function - see Section A.3 - which allows us to relate a surface integral to a volume integral. Applying this to the surface integral for the pressure force, Eq. (4.2), we obtain the relation,

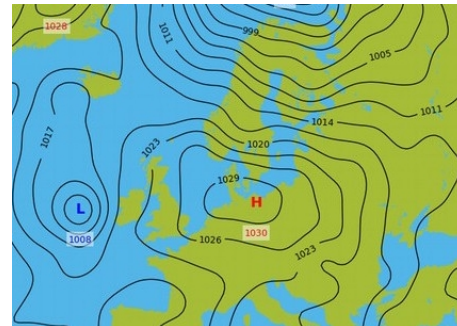
$$- \iint_S P \hat{n} \, dS = - \iiint_V \nabla P \, dV,$$

and can hence write the pressure force as the volume integral in Eq. (4.3).

Gauss' theorem requires that the scalar function (here the fluid pressure) and its derivatives are continuous over the region of interest. Since the fluid pressure can change continuously at a boundary, this result is usually limited to volumes *within the fluid* (away from boundaries).



ASIDE: In the atmosphere, the pressure force causes wind. It is no surprise then that pressure charts are important to meteorologists. The meteorological pressure chart below shows contours of equal atmospheric pressure, termed “isobars”, over Europe and the North Atlantic. Pressure is quoted in units of millibars, with $1 \text{ millibar} = 10^2 \text{ Pa}$. This was a windy day in Scandinavia but fairly still in the UK.



EXAMPLE 14: Consider again Example 13 but now evaluate the pressure force on the cube as a volume integral.

4.3 Body forces

A **body force** is a force experienced throughout the whole volume of a body of fluid.

Body forces are exerted from the outside through external force fields. The most important example to us is gravity, but electric and magnetic forces are other examples.

For later convenience we specify the body force in terms of the body force per unit mass, denoted $\vec{f}$. In general this may be a function of time and position, $\vec{f} = \vec{f}(\vec{x}, t)$, and it has units of acceleration.

The total body force acting on a volume V of fluid with density $\rho = \rho(\vec{x})$ is,

$$\vec{F} = \iiint_V \rho \vec{f} dV. \quad (4.5)$$

It follows that the body force on an infinitesimally-small volume dV is $d\vec{F} = \rho \vec{f} dV$.

Because body forces are proportional to the volume involved, they are classed as *volume forces* (in contrast to the pressure force, which is a *surface force*).

4.3.1 Conservative body forces and potential

A **conservative body force** can be written in the form $\vec{f} = -\nabla\Phi$, where $\Phi = \Phi(\vec{x}, t)^a$ is a scalar field termed the **potential**.

^aThe Greek capital letter “phi”, pronounced “fy”.

For a given conservative body force $\vec{f}$ we can obtain the corresponding potential function Φ via the method of partial integration. This method is described further in Appendix A.5.

As an example of the method, consider $\vec{f} = (2x, 0, 1)$. To obtain a suitable potential Φ which satisfies $\vec{f} = -\nabla\Phi$ we seek a simultaneous solution to the three partial differential equations,

$$\frac{\partial\Phi}{\partial x} = -2x, \quad \frac{\partial\Phi}{\partial y} = 0, \quad \frac{\partial\Phi}{\partial z} = -1.$$

Partial integration gives,

$$\Phi = -x^2 + A(y, z) + c_1, \quad \Phi = B(x, z) + c_2, \quad \Phi = -z + C(x, y) + c_3,$$

for which a suitable overall general solution is,

$$\Phi = -x^2 - z + c,$$

with combined arbitrary constant c .

Due to the presence of c , there are an infinite number of suitable potential functions Φ which vary by an additive constant. It follows that the absolute value of a potential is meaningless, such that:

Only changes in potential are important. For this reason, when stating potentials, we usually set the arbitrary constant to zero.

4.3.2 Gravitational body force

The gravitational body force per unit mass is $\vec{f} = -g\hat{z}$, where $\hat{z}$ is the unit vector along z and $g = 9.8 \text{ m s}^{-2}$ is the acceleration due to gravity.

So the total gravitational body force on a volume V is,

$$\vec{F} = \iiint_V \rho(-g\hat{z}) dV = -g\hat{z} \iiint_V \rho dV = -Mg\hat{z}, \quad (4.6)$$

where M is the mass within V . This force is just the weight[†] of the fluid, as we would expect.

The gravitational body force is conservative and has the **gravitational potential** $\Phi = gz$.

It should be evident that this potential recovers the required gravitational body force $\vec{f} = -g\hat{z}$.

[†] *Weight* is the force acting on a body due to gravity. It is a vector, directed vertically downwards. Often we refer to its magnitude, which is Mg .

4.4 Fluid equilibrium

A fluid is at equilibrium if the pressure forces and body forces cancel everywhere in the fluid according to the **equilibrium condition**,

$$\nabla P = \rho \vec{f}. \quad (4.7)$$

In a fluid at equilibrium, there is no acceleration of any parts of the fluid (since there is no net force acting on any part of the fluid).

Derivation

We derive this by considering some volume V within the fluid. Requiring that there is no net force on the volume, then the body force (4.5) and pressure force (4.2) must balance (sum to zero),

$$\iiint_V \rho \vec{f} dV - \iiint_V \nabla P dV = 0,$$

which we may combine into one integral,

$$\iiint_V (\rho \vec{f} - \nabla P) dV = 0.$$

This condition must be satisfied throughout the whole fluid, no matter what the integration volume, which requires that the integrand must equal zero,

$$\rho \vec{f} - \nabla P = 0,$$

which is the same as Eq. (4.7)

4.5 Incompressible fluid under gravity

In the rest of this chapter we will focus on a static fluid which is (a) incompressible with density ρ_0 , and (b) experiences gravity. This just about covers all *liquids* on Earth.

4.5.1 Solution for the pressure $P(\vec{x})$

The fluid satisfies the equilibrium condition of Eq. (4.7) with $\vec{f} = -g\hat{z}$,

$$\nabla P = -g\rho_0\hat{z}. \quad (4.8)$$

This corresponds to three partial differential equations (one for each component),

$$\frac{\partial P}{\partial x} = 0, \quad \frac{\partial P}{\partial y} = 0, \quad \frac{\partial P}{\partial z} = -g\rho_0. \quad (4.9)$$

These equations tell us that the pressure does not vary in x and y , but does vary in z . We can use the method of partial integration to find the general solution for $P(\vec{x})$,

$$P(x, y, z) = -g\rho_0 z + c.$$

For convenience we define $P(z = 0) = P_0$, leading the following solution.

For a static and incompressible fluid under gravity, the solution for the pressure is,

$$P(z) = P_0 - \rho_0 g z, \quad (4.10)$$

where $P(z = 0) = P_0$ and ρ_0 is the fluid density.

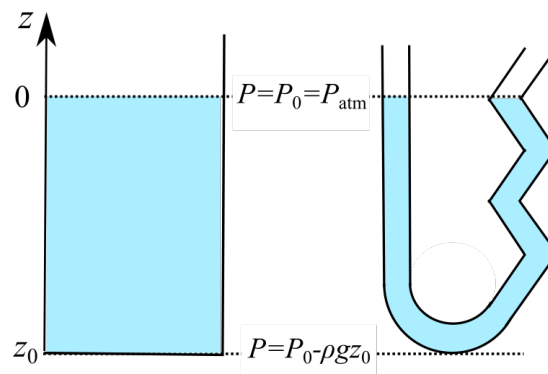
There are several observations to make on this result:

- (i) This solution trivially satisfies the first two differential equations, since $P = P(z)$.
- (ii) The pressure increases with decreasing z (i.e. with depth into the fluid), and does so as a linear function of z .
- (iii) The fact that the pressure does not vary with x and y leads to the statement:

The pressure at a given height in a static and incompressible fluid under gravity^a is uniform.

^aIndeed, this statement is true for any body force which varies only with z .

Thus, in a regular beaker, the pressure is the same across any horizontal plane. It does not matter what the shape of the container is (as long as the fluid is connected), so this also holds for the funny U-shaped container below.



A liquid at equilibrium in two containers, both open to the atmosphere.

- (iv) $z = 0$ might represent the position at some arbitrary vertical level in the fluid, or it might represent some physical boundary, such as the top or bottom of the fluid. For a fluid which is open to the atmosphere, such as a glass of water, it is often convenient to take $z = 0$ as the level of the fluid's surface. We can assume that the air (which is itself a fluid!) is at atmospheric pressure P_{atm} of 10^5Pa . Recall that pressure is continuous throughout any connected fluid system. This means that the top of the fluid must be at atmospheric pressure, in other words that $P_0 = P_{\text{atm}}$.



ASIDE: Standard atmospheric pressure is the average air pressure at sea-level and has the precise value of 101 325 Pa. The pressure at sea-level can vary in time and position - remember, it is the variation of air pressure in space that generates wind - but this change is small (up to a few %).

Atmospheric pressure also changes with height, z , in accordance with Eq. (4.7) but complicated by the fact that the air density can change with z . However, the air pressure varies slowly with altitude, for instance, it changes by less than 1% over 100m. Thus for situations close to the Earth's surface we treat air pressure as constant (taking the value 10^5 Pa for simplicity).



EXAMPLE 15:

- The deepest point in the oceans is the Marianas trench in the Pacific ocean, at a depth of 11 km. What is the pressure here? The pressure at sea level is $P_{\text{atm}} = 10^5$ Pa and the density of the ocean is $\rho_0 = 10^3 \text{ kg m}^{-3}$.
- In the Earth's atmosphere, density and pressure vary with height z and are related as $\rho(z) = P(z)/hg$, where h is a constant ($h \approx 8.2$ km), and g is the gravitational acceleration. Use the equilibrium condition to obtain an expression for $P(z)$.
- Sketch $P(z)$, upwards from the Marianas trench and into the atmosphere.

4.5.2 Transfer of pressure and hydraulic jacks

Imagine that the pressure P_0 at $z = 0$ is suddenly changed to $P_0 + \Delta P$. (If the fluid is in some open container, this could be done, for example, by adding more fluid on top of the fluid or by piston pushing down on the fluid surface.) Then the solution Eq. (4.10) becomes $P(z) = P_0 + \Delta P - \rho_0 g z$. In other words, the pressure *everywhere* increases by ΔP . We can therefore make the statement that:

A pressure change occurring anywhere in an incompressible fluid is transmitted throughout the fluid such that the same change occurs everywhere.

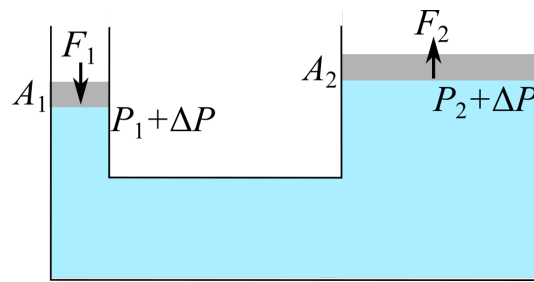
In physics and engineering, this is referred to as *Pascal's law*.

Note that if the fluid is compressible, then the fluid would have another possible behaviour in the presence of a pressure change - to change its density.

Hydraulic jacks

This property is the basis of hydraulic devices. The simplest of these are hydraulic jacks, which are used to lift heavy objects by applying a minimal amount of force.

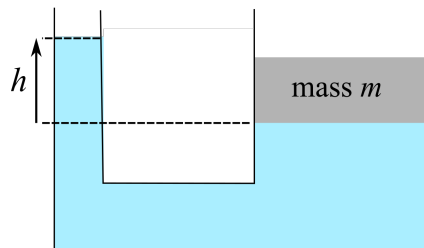
Consider the chamber of incompressible fluid shown, featuring two connected tanks of fluid, each with a piston. A force F_1 is applied downwards on the left piston, which increases the pressure there by $\Delta P = F_1/A_1$, where A_1 is the cross-sectional area. This pressure change is transmitted through the whole fluid, increasing the pressure on the right piston by ΔP . This generates an upward force on the right piston, $F_2 = \Delta P A_2 = F_1 A_2/A_1$. Thus if $A_2 \gg A_1$ the force generated on the right piston is much larger than that applied to the left piston, providing a convenient means of amplifying forces, for example to lift heavy objects.



Schematic of a hydraulic jack.



EXAMPLE 16: Consider a fluid of uniform density in the vessel below. The left side is open to the atmosphere and has a cross-sectional area A_L . The right side experiences a mass m and has cross-sectional area A_R . The fluid is at equilibrium. By considering the pressure on the left and right sides, obtain an expression for the height difference h .



4.5.3 Pressure as the weight of fluid

The pressure at a point within a static fluid is equal to the magnitude of the weight per unit cross-sectional area of the fluid above it^a.

^aThis result holds also for all static fluids under gravity, not just those which are incompressible.

We can see this for an incompressible fluid as follows. Consider some vertical column of incompressible fluid with height h and uniform cross-sectional area A . For simplicity let's say that the pressure at the top is zero. According to Eq. (4.10), the pressure at the bottom will be,

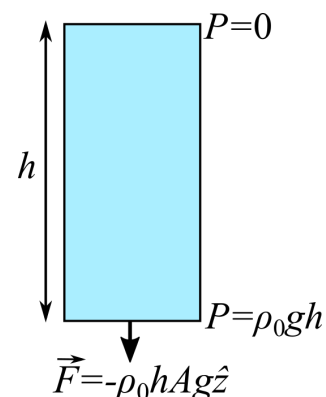
$$P = \rho_0 g h.$$

Meanwhile, the magnitude of the weight of this column of fluid is,

$$Mg = \rho_0 A h g.$$

Thus the pressure is indeed given by the magnitude of the weight per unit cross-sectional area (the weight divided by the cross-sectional area).

So the air pressure acting on you right now is because of the weight of all the air above you, going right up to the top of the Earth's atmosphere. Italian scientist Torricelli aptly described that "we live submerged at the bottom of an ocean of air".



Pressure as weight of fluid.



EXAMPLE 17: Consider now a vertical column within a fluid whose density $\rho(z)$ varies with height z . Repeat the derivation above, but taking into account the variable density, to show that the statement holds.

4.5.4 Buoyancy

It is hard to push a football into water. This is because the water is pushing back with a force known as the *buoyancy force*. It arises for a body immersed in any fluid, under the action of gravity, and arises from pressure forces in the fluid.

A body immersed in a fluid under gravity experiences a **buoyancy force** $\vec{F}_B$ which is equal and opposite to the weight of the displaced fluid, that is,

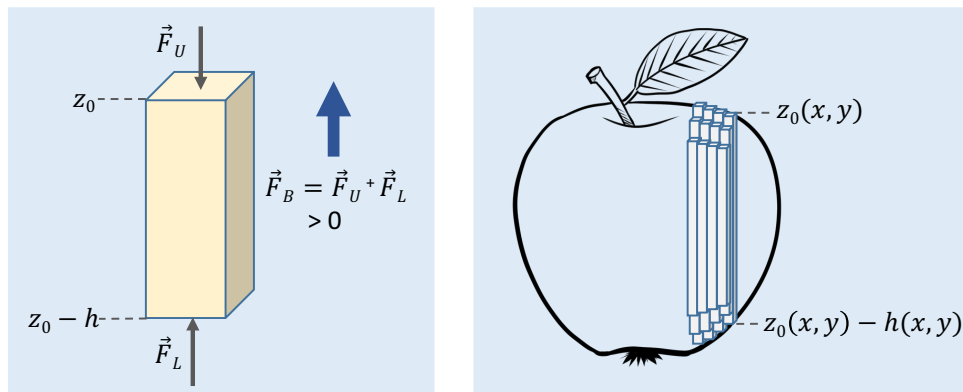
$$\vec{F}_B = M_0 g \hat{z}, \quad (4.11)$$

where M_0 is the mass of the displaced fluid^a.

^aThis result holds for all static fluids, not just those which are incompressible.

Derivation

Consider first a cuboid body submerged in an incompressible fluid under gravity, with height h and cross-sectional area A . The fluid exerts a pressure force on all sides of the cuboid. The pressure forces on the side walls all have the same magnitude, but point in opposite directions and hence cancel out, playing no role on the buoyancy force.



The pressure force on the upper surface acts down and the pressure force on the lower surface acts up. For any fluid under gravity, pressure increases with depth, and so the force on the lower surface will win over that on the upper surface, and hence the net pressure force - the buoyancy force - acts upwards. Let's calculate it. The pressure force on the upper face at $z = z_0$ (taking $\hat{n} = \hat{z}$) is,

$$\vec{F}_U = - \iint_S P \hat{n} dS = -P(z = z_0) \hat{z} \iint_S dS = -P(z = z_0) A \hat{z}$$

Similarly, on the lower face (taking $\hat{n} = -\hat{z}$) it is $\vec{F}_L = P(z = z_0 - h) A \hat{z}$. The buoyancy force is then,

$$\vec{F}_B = \vec{F}_U + \vec{F}_L = (P(z = z_0 - h) - P(z = z_0)) A \hat{z}.$$

Now, for an incompressible fluid under gravity, $P(z) = P_0 - \rho_0 g z$, and so this becomes,

$$\vec{F}_B = \rho_0 g h A \hat{z} = M_0 g \hat{z}.$$

This is equal in magnitude to the weight of the volume if it were filled with the surrounding fluid, and is directed upwards.

If the body is a more general shape, what is the buoyancy force? We treat the body volume as a collection of infinitesimally-narrow cuboids, extending from the top surface at $z = z_0(x, y)$ to the bottom surface at $z = z_0(x, y) - h(x, y)$. (This is the 3D analog integrating the 2D area under a curve as a collection of infinitesimally-narrow rectangles.)

According to the result above, the buoyancy force on one such cuboid, centred at (x, y) , is given by,

$$d\vec{F}_B = \rho_0 g h(x, y) \hat{z} dS,$$

where we have replaced the cross-sectional area A with the infinitesimal area dS . The total buoyancy force on all such cuboids is given by integrating (summing) all of these contributions over the whole upper surface,

$$\vec{F}_B = \iint_S d\vec{F}_B = \rho_0 g \hat{z} \iint_S h(x, y) dS.$$

The integral is simply the volume of the body V , and since $M_0 = \rho_0 V$ we finally obtain the expected result,

$$\vec{F}_B = M_0 g \hat{z}.$$



ASIDE: The fact that the buoyancy force is equal and opposite to the weight of the body can also be argued as follows. Imagine that the body is itself a blob of the fluid. If the whole fluid is static, then this blob must have zero net force acting on it (otherwise it would accelerate in some direction). We know that this blob has a downward weight due to gravity, and so for the net force on it to be zero there must exist an equal and opposite force to the blob's weight which acts upwards.

Sink or float?

If the body has a larger density than the displaced fluid, the body will tend to **sink**, while if the body has a lower density it will tend to **rise**. If the densities are the same the body is **neutrally buoyant**.

We can understand this by considering the net force on the body,

$$\vec{F}_{\text{net}} = \text{weight force} + \text{buoyancy force} = g \hat{z} V (\rho_0 - \rho_b). \quad (4.12)$$

where ρ_b is the density of the body. If $\rho_b > \rho_0$, the net force is downwards (gravity beats buoyancy) and the body will sink. If $\rho_b < \rho_0$ then the net force is upwards (buoyancy beats gravity) and the body will rise.

The definition of the buoyancy force in terms of the weight of displaced fluid holds even for a body which crosses the boundary of two fluids. An important example of this is a body floating at the surface of water, where it is partially displacing water and partially displacing air. However, in this case, the contribution to the buoyancy force from air is usually so small compared to that from the water that we can ignore it (due to the vastly greater density of water compared to air; water is 1000 times more dense).



ASIDE: The buoyancy force is the reason we feel like we weight less when we are in a swimming pool. Whereas in air, our net downward force is our weight (ignoring the negligible buoyancy effect from air), when we are in water our net downward force is the net of our true weight and the opposing buoyancy force. If we were to stand of a weighing scales in the pool, the scales would measure this net force, making it look like we weighed less!



EXAMPLE 18: Estimate the buoyancy force acting on a typical person due to the surrounding air at sea level. You may take the density of air to be 1.2 kg m^{-3} and the density of the human body to be the same as water, 1000 kg m^{-3} .



EXAMPLE 19: A cube of uniform density ρ_b and sides of length L floats at the surface of water. A height h of the object lies above the water in the air. The density of water and air are denoted ρ_w and ρ_a , respectively. You may consider the cube to be in equilibrium with the fluid.

- By considering the forces on the cube, derive an expression for the fraction of the cube's height which is above the water. You may ignore the contribution to buoyancy from the air.
- Apply your answer to estimate what fraction of an iceberg's height is above the water. The density of ice is 917 kg/m^{-3} and the density of sea water is 1030 kg/m^{-3} .
- Now consider the buoyancy force arising from the displaced air. Determine an expression for the ratio of the buoyancy force from the displaced air to the buoyancy force from the displaced water. Evaluate this ratio for the iceberg scenario, taking $\rho_a = 1 \text{ kg/m}^{-3}$.



ASIDE: Archimedes (287-212BC) was a Greek polymath, famous for his studies of objects in water. He was the first to formulate the idea of a buoyancy force - that a body immersed in a fluid is buoyed up by a force equal in magnitude to the weight of dispaced fluid. This is often termed "Archimedes principle".

This discovery was made as Archimedes helped the King of Syracuse, who suspected that his gold crown was a forgery, made of silver instead. However, the king could not check this since he could not measure its density; the shape of the crown was too complicated to measure its volume. The king asked Archimedes for help. While taking a bath, Archimedes realised that an object in water displaces the same volume of water as the object. This led Archimedes to crying his famous "Eureka" (supposedly he jumped out of the bath and ran out into the street naked to do this).

By dividing the mass of the crown by the volume of water displaced, he found the density of the crown was consistent with silver, not gold. The crown was a forgery.

5 Inviscid fluids under motion

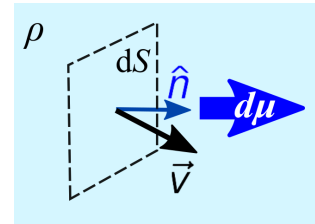
We will next introduce the equations that describe the *motion* of an *inviscid fluid*, for a flow velocity $\vec{v} = \vec{v}(\vec{x}, t)$ and density $\rho = \rho(\vec{x}, t)$ which may vary in time and position.

5.1 Mass flux

The **mass flux** μ^a is the rate at which mass flows through some surface. The infinitesimal mass flux $d\mu$ through a surface element dS , in the direction of the normal unit vector $\hat{n}$, is,

$$d\mu = \rho \vec{v} \cdot \hat{n} dS.$$

^aThe Greek letter “mu”.



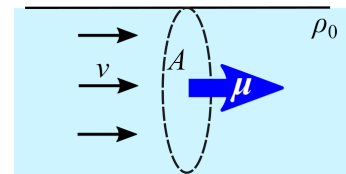
Infinitesimal mass flow rate $d\mu$.

The mass flux is often termed the “mass flow rate”. From this definition, we can write the total mass flux through a surface S as the integral,

$$\mu = \iint_S \rho \vec{v} \cdot \hat{n} dS.$$

The mass flux takes a simple form for an incompressible fluid of density ρ_0 undergoing uniform flow. As an example, consider such flow along a pipe at *speed* v . If the surface S is a cross-section of the pipe, then $\vec{v} \cdot \hat{n} = v$, and the mass flux is,

$$\mu = \rho_0 v \iint_S dS = \rho_0 v A,$$



Mass flux μ down a pipe.

where A is the cross-sectional area of the pipe.

The *volumetric flux*, or *volumetric flow rate*, is closely related to the mass flux. It is the rate at which volume of fluid (rather than mass of fluid) flows through S . It is the mass flux divided by density.

5.2 Conservation of mass

The law of **conservation of mass** says that mass cannot be spontaneously created or destroyed.

Thus, if fluid flows in (or out) of some region, the mass of fluid within the region must increase (or decrease) accordingly. For example, consider a bathtub of water, whose tap is running. The amount of water within the bathtub increases with time. By the conservation of mass, this increase must equal the amount of water passing through the tap. We can formulate this idea in general as:

For a volume V of fluid mass M , the conservation of mass requires that,

$$\frac{dM}{dt} = \text{total mass flux into } V. \quad (5.1)$$

If the mass flux is *out of* V then it is negative. There may be multiple contributions to the total mass flux, for example, if the bath plughole is open while the tap is on. We will meet such scenarios shortly.

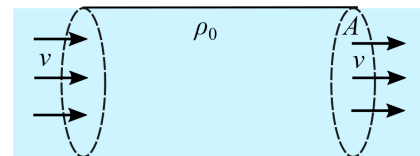
5.2.1 Incompressible fluid

For an incompressible fluid, mass conservation is relatively straightforward to apply. Let's consider some examples.

Uniform pipe flow

Consider a pipe of fixed cross-sectional area A along which an incompressible fluid undergoes steady and uniform flow at speed v . Picture a section of the pipe with volume V , as shown.

The flux into V through the left surface is $\rho_0 v A$ and through the right surface it is $-\rho_0 v A$. The net mass flux into V is thus zero and so we can write,



Uniform mass flow down a pipe.

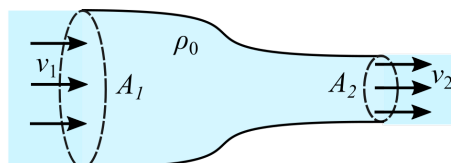
$$\frac{dM}{dt} = 0$$

Does this result make sense? Since the volume and the fluid density are fixed, the mass of the fluid within V must also be constant over time, giving the same result.

If the mass flux were not zero, then this would have to be accommodated either by a change in the fluid density or the pipe volume, both of which are not allowed for here.



EXAMPLE 20: Consider again steady and uniform flow through a section of pipe, but now where the cross-sectional area changes as shown below.



The flow is steady and enters at speed v_1 , where the area is A_1 . How fast is the fluid at the exit, v_2 , where the area is A_2 ? Take the fluid to be incompressible.

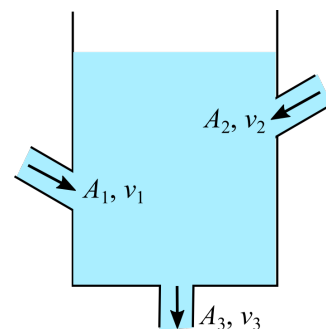
Tank with inlets and outlets

Now consider a tank of incompressible fluid in which fluid enters the tank at two inlets and exits at an outlet. How does the fluid mass change with time?

The total mass flux into the tank is $\rho_0 A_1 v_1 + \rho_0 A_2 v_2 - \rho_0 A_3 v_3$, and so Equation (5.1) gives,

$$\frac{dM}{dt} = \rho_0 (A_1 v_1 + A_2 v_2 - A_3 v_3).$$

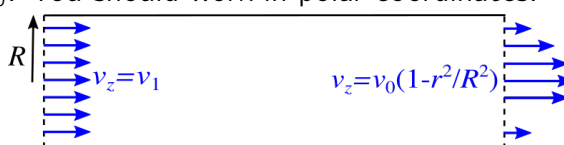
If the total mass flux is negative (the outflow wins), M will decrease with time. If it is positive (the inflow wins), M will increase with time. If it is zero (the outflow is exactly matched by the inflow), M stays constant over time.



Tank with 2 inlets and 1 outlet.



EXAMPLE 21: Consider water (an incompressible fluid) flowing along a cylindrical pipe of radius R , which lies along the z axis. The water is forced into the pipe with uniform velocity $\vec{v} = (0, 0, v_1)$, with v_1 a constant. By the end of the pipe, the flow adopts a curved flow profile $\vec{v} = (0, 0, v_0(1 - r^2/R^2))$, where r is the radial coordinate^a. By applying mass conservation, deduce a relation between v_1 and v_0 . You should work in polar coordinates.



^aThis is a more realistic flow profile for water passing down a pipe since the friction between the water and the pipe tends to cause the water to become stationary at the pipe wall.

5.3 Continuity equation

How do we apply conservation of mass to a more general fluid where the density and velocity may vary? Here we present some general equations for this.

For a general fluid in some volume V , enclosed by the surface S , the conservation of mass is expressed through the **continuity equation (integral form)**,

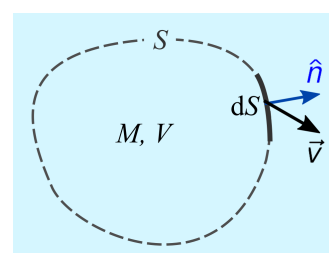
$$\iiint_V \frac{\partial \rho}{\partial t} dV = - \iint_S \rho \vec{v} \cdot \hat{n} dS. \quad (5.2)$$

Derivation

We apply Eq. (5.1) to a general fluid of varying density and velocity. The fluid mass M within V is $M(t) = \iiint_V \rho(\vec{x}, t) dV$,

such that we can express dM/dt as,

$$\frac{dM}{dt} = \frac{d}{dt} \iiint_V \rho dV = \iiint_V \frac{\partial \rho}{\partial t} dV. \quad (5.3)$$



Mass flux for a closed surface S .

Meanwhile, the total mass flux *in* through the surface S is $-\iint_S \rho \vec{v} \cdot \hat{n} dS$. Inserting these results into Eq. (5.1) gives

the continuity equation (integral form),

$$\iiint_V \frac{\partial \rho}{\partial t} dV = - \iint_S \rho \vec{v} \cdot \hat{n} dS.$$



EXAMPLE 22: A fluid has density $\rho(\vec{x}) = x$ and velocity $\vec{v}(\vec{x}) = (v_x, v_y, v_z) = (y/x, -1, 0)$. Verify that the continuity equation is satisfied over a cubic volume $0 \leq x, y, z \leq 1$ by evaluating both sides of the equation.

5.3.1 Continuity equation - differential form

For a volume within the fluid, the continuity equation can be written in the **differential form**,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0. \quad (5.4)$$

Derivation

We use Gauss' theorem - see Section A.3 - to express the mass flux surface integral in Eq. (5.2) as the volume integral of the divergence of the mass flux, that is,

$$\iint_S \rho \vec{v} \cdot \hat{n} dS = \iiint_V \nabla \cdot (\rho \vec{v}) dV.$$

Then Eq. (5.2) becomes,

$$\iiint_V \frac{\partial \rho}{\partial t} dV = - \iiint_V \nabla \cdot (\rho \vec{v}) dV, \quad \text{i.e.} \quad \iiint_V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) \right] dV = 0. \quad (5.5)$$

This must hold over any volume V within the fluid, which requires that the integrand must vanish at every point in space. We can then drop the integration, leading to Eq. (5.4).



EXAMPLE 23: If the fluid density is $\rho = e^{-x^2-y^2-z^2-\alpha t}$ and the fluid velocity is $\vec{v} = (v_x, v_y, v_z) = (-y, x, 0)$, determine the value/s of α for which the continuity equation (differential form) is satisfied.



EXAMPLE 24: For a fluid of uniform density ρ_0 , the rate at which the fluid mass travels out of a volume V satisfies Gauss' theorem,

$$\rho_0 \iiint_V \nabla \cdot \vec{v} dV = \rho_0 \iint_S \vec{v} \cdot \hat{n} dS.$$

Consider a flow velocity $\vec{v} = (v_r, v_\theta, v_z) = (r^2, 2, 3z)$. Verify that this result holds over a volume corresponding to the cylinder $0 \leq z \leq 1$, $0 \leq r \leq 1$, by evaluating both sides of the equation. Work in cylindrical polar coordinates, using the form of the divergence provided in Appendix A.7.

5.4 Incompressible flows

An **incompressible flow** is a flow in which a fluid parcel remains at a uniform density as it moves.

It is subtly different, and more general than, the flow of an incompressible fluid. An incompressible fluid always undergoes incompressible flow. However a fluid undergoing incompressible flow does not have to be an incompressible fluid...the flow behaviour can be such that the density of a fluid parcel stays fixed as it moves, even though the fluid as a whole has variable density.

A flow is incompressible if it satisfies the **incompressibility condition**,

$$\nabla \cdot \vec{v} = 0. \quad (5.6)$$

Derivation

The fluid flow must satisfy the differential form of the continuity equation $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$. Using the product rule to expand the $\nabla \cdot (\rho \vec{v})$ term gives,

$$\frac{\partial \rho}{\partial t} + \rho(\nabla \cdot \vec{v}) + \vec{v} \cdot (\nabla \rho) = \frac{D\rho}{Dt} + \rho(\nabla \cdot \vec{v}) = 0, \quad (5.7)$$

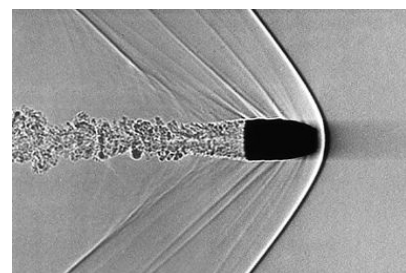
where we have introduced the material derivative.

We require that the density of the fluid parcel does not change as it moves with the fluid, which we can write as $\frac{D\rho}{Dt} = 0$. Then above equation reduces to $\rho(\nabla \cdot \vec{v}) = 0$. Since the density must be non-zero for a fluid parcel to exist, this result simplifies further to $\nabla \cdot \vec{v} = 0$, the incompressibility condition.



ASIDE: Air is not an incompressible fluid since its density can be readily made to change by changing the pressure it experiences. However, air closely approximates incompressible flow providing the flow speed v remain less than the speed of sound in air, c . We can express this condition as $Ma < 1$, where Ma is the “Mach number”, v/c . The speed of sound in air is around 340 m s^{-1} . The flow of air past a car below is an example of incompressible flow.

If the Mach number is comparable to or exceeds 1 then the fluid cannot move out of the way quickly enough for fluid parcels to stay at constant density and the fluid bunches up in places. An example is the bullet below, which forms a high-density shock wave in front of it.





EXAMPLE 25: A flow is described by the velocity field $\vec{v} = (\alpha x, \beta y, \gamma z)$. What relation must exist between the coefficients α , β and γ for the flow to be incompressible?

5.5 The Euler equation

The flow of an inviscid (but otherwise general) fluid under the influence of a body force is described by the **Euler equation**,

$$\frac{D\vec{v}}{Dt} = -\frac{1}{\rho}\nabla P + \vec{f}, \quad (5.8)$$

The right side represents the net force on the fluid. If this is zero, then the fluid does not accelerate, and this equation reduces to that of fluid equilibrium - Eq. (4.7).

Derivation

Consider the motion of a fluid parcel, of volume dV , moving with the flow, under the influence of a body force $\vec{f}$. We apply Newton's second law " $F=ma$ " to the fluid parcel, treating it like an object which accelerates subject to the net forces it experiences. The net force on the parcel is the sum of the pressure force $-\nabla P dV$ and the body force $\rho \vec{f} dV$.

The mass of the parcel is ρdV and its acceleration is $\vec{a} = \frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v}$. Applying $m\vec{a} = \vec{F}$ gives,

$$\rho dV \frac{D\vec{v}}{Dt} = -\nabla P dV + \rho \vec{f} dV. \quad (5.9)$$

Cancelling dV and dividing through by ρ gives the above Euler equation,

$$\frac{D\vec{v}}{Dt} = -\frac{1}{\rho}\nabla P + \vec{f}. \quad (5.10)$$

5.6 The incompressible Euler equation

For the rest of this chapter we will focus on *incompressible fluids* with a *conservative body force*. Under these assumptions the Euler equation takes a simplified form. This model is particularly useful for describing water (an incompressible fluid) on Earth (gravity being a conservative body force).

For an incompressible fluid with a conservative body force, the Euler equation reduces to the **incompressible Euler equation**,

$$\frac{\partial \vec{v}}{\partial t} + \vec{\omega} \times \vec{v} = -\nabla B, \quad (5.11)$$

where $B = \frac{P}{\rho} + \frac{1}{2}v^2 + \Phi$ is the **Bernoulli function**.

Derivation

Starting from the Euler equation (5.8) we introduce that the fluid is incompressible (ρ constant) and the force is conservative ($\vec{f} = -\nabla\Phi$), and manipulate,

$$\begin{aligned} \frac{D\vec{v}}{Dt} &= -\frac{1}{\rho}\nabla P + \vec{f} = -\nabla\left(\frac{P}{\rho}\right) - \nabla\Phi = -\nabla\left(\frac{P}{\rho} + \Phi\right), \\ \text{or } \frac{\partial\vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} &= -\nabla\left(\frac{P}{\rho} + \Phi\right). \end{aligned} \quad (5.12)$$

Next we employ the vector identity $(\vec{A} \cdot \nabla)\vec{A} = (\nabla \times \vec{A}) \times \vec{A} + \nabla(\frac{1}{2}A^2)$, where $A = |\vec{A}|$ to expand out the $(\vec{v} \cdot \nabla)\vec{v}$ term to give,

$$\frac{\partial\vec{v}}{\partial t} + (\nabla \times \vec{v}) \times \vec{v} + \nabla\left(\frac{1}{2}v^2\right) = -\nabla\left(\frac{P}{\rho} + \Phi\right), \quad (5.13)$$

$$\rightarrow \frac{\partial\vec{v}}{\partial t} + \vec{\omega} \times \vec{v} = -\nabla\left(\frac{P}{\rho} + \frac{1}{2}v^2 + \Phi\right), \quad (5.14)$$

where we have introduced the vorticity $\vec{\omega} = \nabla \times \vec{v}$.



EXAMPLE 26: An incompressible fluid of density ρ_0 under gravity undergoes solid-body rotation, $\vec{v} = (v_x, v_y, v_z) = (-\Omega y, \Omega x, 0)$.

- Starting from the incompressible Euler equation, obtain 3 differential equations for $\frac{\partial P}{\partial x}$, $\frac{\partial P}{\partial y}$ and $\frac{\partial P}{\partial z}$.
- Using the method of partial integration, integrate each equation and thereby construct an overall solution for the fluid pressure $P(\vec{x})$, setting $P_0 = P((0, 0, 0))$.
- Find an equation $z(x, y)$ for surfaces of constant pressure.

Note: this question applies to a bucket of water which is under steady rotation. At the water's surface, the fluid pressure is atmospheric pressure, a constant. So in part (c) you have actually worked out the shape of the water's surface!

5.7 Vorticity equation

The vorticity $\vec{\omega}$ of an incompressible fluid follows the **vorticity transport equation**,

$$\frac{D\vec{\omega}}{Dt} = (\vec{\omega} \cdot \nabla)\vec{v}. \quad (5.15)$$

Derivation

We take the curl of both sides of the incompressible Euler equation, Eq. (5.11),

$$\nabla \times \frac{\partial\vec{v}}{\partial t} + \nabla \times (\vec{\omega} \times \vec{v}) = -\nabla \times \nabla B.$$

It does not matter which order we do derivatives in, and since curl is just a form of derivative we can move it within the $\partial/\partial t$. On the right side we have the curl of a gradient,

which is zero. Then we expand the remaining term using the identity $\nabla \times (\vec{C} \times \vec{D}) = (\vec{D} \cdot \nabla)\vec{C} - (\vec{C} \cdot \nabla)\vec{D} + \vec{C}(\nabla \cdot \vec{D}) - \vec{D}(\nabla \cdot \vec{C})$ to give,

$$\frac{\partial \vec{\omega}}{\partial t} + (\vec{v} \cdot \nabla)\vec{\omega} - (\vec{\omega} \cdot \nabla)\vec{v} + \vec{\omega}(\nabla \cdot \vec{v}) - \vec{v}(\nabla \cdot \vec{\omega}) = 0.$$

The flow is incompressible so $\nabla \cdot \vec{v} = 0$. Also $\nabla \cdot \vec{\omega} = 0$ since the gradient of a curl is always zero. We then obtain,

$$\frac{\partial \vec{\omega}}{\partial t} + (\vec{v} \cdot \nabla)\vec{\omega} = (\vec{\omega} \cdot \nabla)\vec{v}, \quad \text{or} \quad \frac{D\vec{\omega}}{Dt} = (\vec{\omega} \cdot \nabla)\vec{v}.$$

If the flow is also 2D, then the vorticity of each fluid parcel is conserved as it moves with the flow.

To see this, we recall that for a 2D flow $\vec{v} = (v_x, v_y, 0)$ the vorticity has the form $\vec{\omega} = (0, 0, \omega_z)$, for which the right side of the vorticity transport equation becomes,

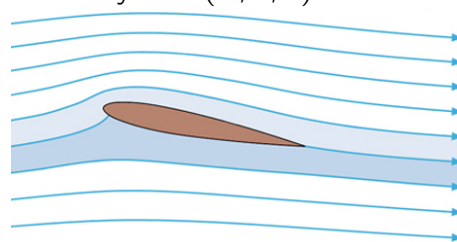
$$\begin{aligned} \frac{D\vec{\omega}}{Dt} &= (\vec{\omega} \cdot \nabla)\vec{v} \\ &= \omega_z \frac{\partial \vec{v}}{\partial z} \\ &= \vec{0}. \end{aligned}$$

This directly implies that the vorticity is conserved for each fluid parcel.



EXAMPLE 27:

- (a) If the 2D flow of an incompressible fluid is additionally *steady*, show that $\vec{\omega}$ is constant along a streamline.
- (b) Consider 2D flow past the cross-section of a wing. The flow is steady and the fluid can be taken to be incompressible. The flow originates far to the left with uniform velocity $\vec{v} = (\alpha, 0, 0)$. What is the vorticity here?



- (c) Using parts (a) and (b), explain why the whole flow must be irrotational.

5.8 Bernoulli's theorem

For steady flow of an incompressible fluid, the flow follows *Bernoulli's theorem*, the most widely used relation in fluid dynamics. It is simple to use and gives useful insight into the balance of pressure, fluid velocity and potential (or elevation, in the case of gravitational potential) in a fluid.

Bernoulli's theorem states that, for steady flow of an incompressible fluid, the Bernoulli function $B = \frac{P}{\rho} + \frac{1}{2}v^2 + \Phi$ is constant along a streamline.

Bernoulli's theorem is important to allow us to relate the properties of the flow along a streamline. Note that the value of B may be different on different streamlines.

Derivation

For a steady flow, the incompressible Euler equation (5.11) reduces to,

$$\vec{\omega} \times \vec{v} = -\nabla B. \quad (5.16)$$

On taking the dot product with $\vec{v}$ we obtain,

$$\begin{aligned} \vec{v} \cdot (\vec{\omega} \times \vec{v}) &= \vec{v} \cdot (-\nabla B) \\ &= -(\vec{v} \cdot \nabla)B. \end{aligned} \quad (5.17)$$

Using the identity that $\vec{C} \cdot (\vec{D} \times \vec{C}) = 0$, for general vectors $\vec{C}$ and $\vec{D}$, we obtain,

$$(\vec{v} \cdot \nabla)B = 0, \quad (5.18)$$

which implies that B is constant along a streamline, as stated in Bernoulli's theorem.



ASIDE: The Bernoulli function has a physical meaning: it represents the energy of the fluid parcel. $\frac{P}{\rho}$ is the “internal” energy, that is, the energy associated with the random motion of the microscopic particles. $\frac{1}{2}v^2$ is the kinetic energy of the moving fluid. Φ is the potential energy of the fluid, i.e. the energy it has by way of its position in a force field such as gravity. Bernoulli's theorem thus tells us that the energy of the fluid parcel is conserved as it moves along a streamline. For example, if the fluid parcel's kinetic energy increases, its internal and/or potential energy must decrease accordingly.



EXAMPLE 28: Water is supplied to a building from a main water pipe just under the road outside. Assume that the flow is steady and that the main pipe is so large in diameter that the flow speed is small enough to be taken as zero. The pressure in the main pipe is 10^6 Pa, the density of water is 10^3 kg/m³ and atmospheric pressure is $P_{\text{atm}} = 10^5$ Pa. Using the Bernoulli theorem:

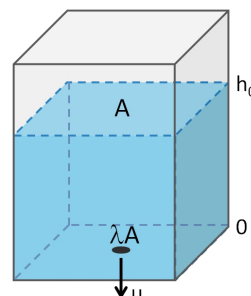
- Estimate how high the water can rise in the building, by considering a streamline from the main pipe up an open vertical pipe.
- Estimate how fast will the water come out of a tap close to ground level, by considering a streamline from the main pipe to the open tap. Give your answer in km/hour.

*This identity makes sense geometrically: $\vec{D} \times \vec{C}$ is a vector that is perpendicular to $\vec{C}$. Thus if we dot it with $\vec{C}$ then answer must be zero (the dot product between two perpendicular vectors is zero).



EXAMPLE 29: A tank of water (of uniform density ρ_0), open to the atmosphere, drains at speed u through a hole in its base. The height of water is $h(t)$. The tank has cross-sectional area A and the hole has cross-sectional area λA , with $\lambda \ll 1$.

- Apply mass conservation to relate the rate of change of fluid mass in the tank to the mass flux through the hole. Hence show that $\frac{dh}{dt} = -\lambda u$.
- Use Bernoulli's theorem to show that $u = \sqrt{2gh}$ (*Torricelli's law*). You may assume that the speed of fluid at the surface is small enough to be taken to be zero.
- By combining the answers from (a) and (b), form a differential equation of the form $\frac{dh}{dt} = f(h)$. Solve it to find $h(t)$, if the initial height of water is h_0 .
- Obtain an expression for the time it takes for the tank to empty.



5.8.1 Irrotational Bernoulli's theorem

If the flow is also irrotational, then the **irrotational Bernoulli's theorem** states that the Bernoulli function $B = \frac{P}{\rho} + \frac{1}{2}v^2 + \Phi$ is constant everywhere throughout the fluid.

We immediately see this by inserting $\vec{\omega} = 0$ into Eq. (5.16), which leads to $\nabla B = 0$. This implies that B is constant in space.

This result is more powerful than Bernoulli's theorem since it allows us to relate the fluid properties between any two points in the fluid (irrespective of whether they are connected by a streamline).

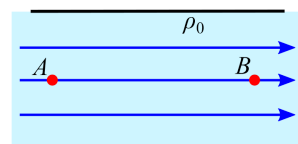


EXAMPLE 30: Consider a fluid of uniform density which flows as a free vortex $\vec{v} = (v_r, v_\theta, v_z) = (0, 1/r, 0)$. Gravity acts. Use the irrotational Bernoulli's theorem to obtain an expression for how the pressure varies with r and z . Where does the pressure reach its maximum and minimum values?

5.8.2 The Bernoulli effect

Under Bernoulli's theorem and for constant potential, regions of higher flow speed have reduced fluid pressure.

For example, consider the steady flow of an incompressible fluid along a horizontal pipe. For two points A and B on a horizontal



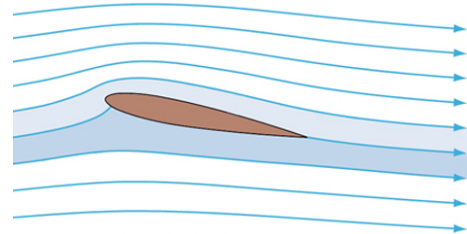
streamline we can use Bernoulli's theorem to state that,

$$\frac{P_A}{\rho_0} + \frac{1}{2}v_A^2 = \frac{P_B}{\rho_0} + \frac{1}{2}v_B^2. \quad (5.19)$$

Thus if the speed is higher at A then the pressure must be less there. The effect can also arise if the streamline is not horizontal but then the varying gravitational potential must be taken into account.



ASIDE: The most important example of the Bernoulli effect is in flight. Consider the steady flow of air past a wing^a. The air flows faster past the top of the wing than the bottom^b. By the Bernoulli effect, the pressure at the top of the wing will be lower than at the bottom. This pressure imbalance creates a net force on the wing acting perpendicular to the overall flow. This force - *lift* - is responsible for keeping planes aloft.



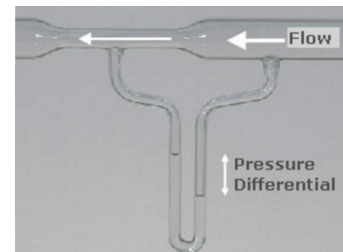
^a The flow is steady for shallow angles of attack (the wing close to horizontal). For larger angles, the flow becomes messy and time-dependent.

^b The reason for this is complicated and beyond our scope, so we just quote the fact.



EXAMPLE 31:

A *venturi meter* is a simple device to measure the speed of an incompressible fluid along a pipe. Consider the fluid to have uniform density ρ_0 . The main flow is at speed v_1 , pressure P_1 and has cross-sectional area A_1 . The meter consists of a constriction, where the flow speed is v_2 , pressure is P_2 and cross-sectional area is A_2 . Using mass conservation and Bernoulli's theorem, show that the main flow speed v_1 can be expressed in terms of the areas, fluid density and pressure difference.



In practice, the pressure difference is visualised by connecting a U-shaped pipe to both regions which contains some fluid; the pressure difference is proportional to the difference in height between the levels of this fluid.

5.8.3 Stagnation pressure

For steady flow past a wing (above) some streamlines pass over the wing and some under it, but there is a special streamline which divided these two behaviours, intercepting the nose of the wing. This *stagnation streamline* arises whenever a fluid flows past any hard, blunt object. It intercepts the surface at normal incidence and the velocity along the streamline must *stagnate* at the surface (since the fluid can't flow into the wing).

The **stagnation pressure** P_s is the fluid pressure at a stagnation point of the flow.

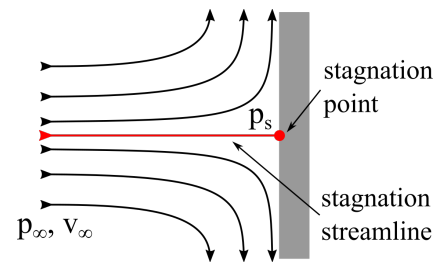
Consider the simpler set-up of a fluid incident upon a wall. Far to the left, the fluid has pressure P_∞ and uniform speed v_∞ . The stagnation streamline is horizontal in this setup. Equating the Bernoulli function along this streamline, at infinity and the stagnation point, gives,

$$\frac{P_\infty}{\rho_0} + \frac{1}{2}v_\infty^2 = \frac{P_s}{\rho_0}. \quad (5.20)$$

Rearranging gives an expression for the stagnation pressure,

$$P_s = P_\infty + \frac{1}{2}\rho_0 v_\infty^2. \quad (5.21)$$

It is the highest pressure found anywhere in the flow field.



5.8.4 The Pitot tube

An important application of Bernoulli's theorem is the Pitot tube, a simple device for measuring the flow speed of fluids. It is most famously used on a plane to measure the speed of air flowing past a plane and hence the plane's speed. It is L-shaped and sticks out of the fuselage of the plane, pointing directly upstream into the flow, as shown in Fig. 5.1. At the tip of the tube (point A), the flow stagnates and has the stagnation pressure P_s . Down the side of the tube (point B), the flow is representative of the main flow, akin to P_∞ above. The pressure difference between these two points is measured.

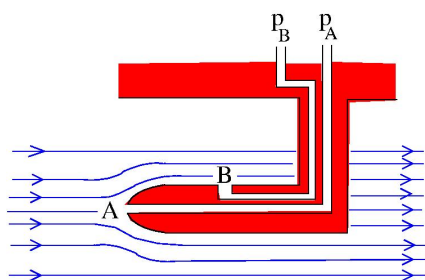


Figure 5.1: An illustration of a pitot tube and a photo of one on the fuselage of a plane.

Bernoulli's theorem applies since we can imagine points A and B are approximately connected by a streamline which goes from very close to point A to the point B. Rearranging Eq. (5.21) we see that the fluid velocity of the main flow is,

$$v = \sqrt{\frac{2(P_s - P_\infty)}{\rho_0}},$$

and can hence be determined from the measured pressure difference between points A and B, and the known density of air.



EXAMPLE 32: Consider the 2D flow $\vec{v} = (x, y)$ of an incompressible fluid with density ρ_0 . Gravity acts on the fluid, and y is taken to be the vertical direction.

- Find the streamline $y(x)$ passing through the point $(1, 2)$.
- Use Bernoulli's theorem to determine how the pressure varies with x along the streamline in (a), if the pressure at the point $(1, 2)$ is P_{12} .
- Identify the stagnation point and the stagnation pressure.

6 Irrotational and incompressible flow

In this chapter we focus on flows which are (i) irrotational and (ii) incompressible. Under these assumptions, the flow solutions can be predicted using an elegant method known as “potential theory”. Many important flow scenarios can be considered. For example, this theory can provide us with the flow velocity around objects of arbitrary shape. From this, we can go on to calculate the pressure distribution (using the irrotational Bernoulli’s theorem) and hence the forces on the object. This has important uses in engineering for designing building, supports and vehicles.

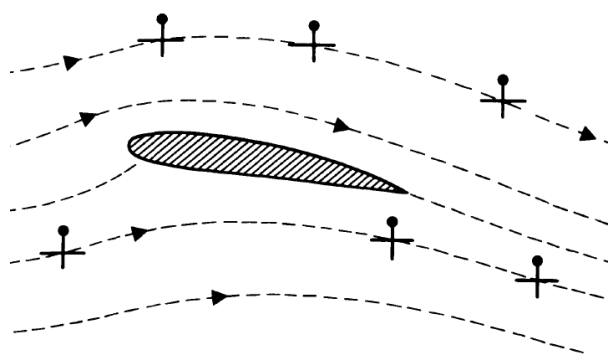
However, the most important application of this method in fluid dynamics, and the one which drove its mathematical development, is flow past a wing. The figure below illustrates the flow of air past a cross-section of a wing. Note how the wing is taken to be stationary with air flowing past it; this is entirely equivalent to a wing moving through otherwise stationary air but is easier to set-up mathematically (since the boundary conditions at the wing surface do not change with time). The flow has two key properties:

- It is incompressible, providing the flow speed remains relatively low as discussed in Section 5.4);
- It is irrotational, as we argued in Example 27 and as depicted in the figure below.

Hence the flow past a wing can be treated with potential theory.

This theory is far from perfect however. It fails for high flow speeds, close to surfaces, and for obstacles whose shape changes rapidly in space. In these cases, viscosity and vorticity play a significant role and one must use a more advanced theory. This is the focus of the stage 3 module MAS3803 Advanced Fluid Dynamics.

For simplicity we will only consider 2D flows in this chapter.



Steady 2D flow of air past a cross-section of a wing. Dashed lines show streamlines. The flow is irrotational: the crosses do not rotate as they move with the flow. Image taken from “Elementary Fluid Dynamics” by D. J. Acheson.

6.1 Velocity potential and potential flow

The **velocity potential** $\phi(\vec{x})$ is defined such that $\vec{v} = \nabla\phi$.

It is convenient to represent the velocity field using the potential function, since it reduces the problem from having two unknowns (v_x and v_y , for the case of 2D flow) to only one unknown (ϕ).

Recall that an irrotational flow is one for which the vorticity $\vec{\omega} = \nabla \times \vec{v} = 0$. In defining $\vec{v}$ as the gradient of a scalar function, the flow is guaranteed to be irrotational since the curl of a gradient is always zero:

$$\vec{\omega} = \nabla \times \vec{v} = \nabla \times \nabla\phi = 0.$$

Such flow is often called **potential flow**.

In Cartesian (x, y) coordinates and cylindrical polar coordinates (r, θ) , we write $\vec{v} = \nabla\phi$ as,

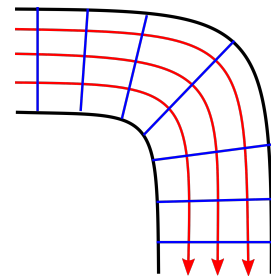
$$\vec{v} = (v_x, v_y) = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y} \right), \quad \vec{v} = (v_r, v_\theta) = \left(\frac{\partial\phi}{\partial r}, \frac{1}{r} \frac{\partial\phi}{\partial \theta} \right). \quad (6.1)$$

Given $\vec{v}(\vec{x})$, we can construct a suitable velocity potential $\phi(\vec{x})$ by partial integration (see Section 4.3.1) of $\nabla\phi$. When doing so, we tend to ignore the arbitrary constant of integration, as we did for potentials of conservative body forces in Section 4.3.1.

6.1.1 Potential lines

Lines of constant velocity potential, known as **potential lines**, are perpendicular to streamlines.

We saw this relationship in the last example. A schematic of the relationship between streamlines and potential lines is shown for flow along a curved pipe. Lines of constant $\phi(x, y)$ satisfy the implicit relation $\phi(x, y) = C$, where C is a constant. We can visualise the velocity potential $\phi(x, y)$ as a landscape in the xy plane; the potential line is a particular contour of this landscape at height C .



Streamlines (red) and potential lines (blue) for flow along a curved pipe.

Derivation

Implicit differentiation of the multivariate function $\phi(x, y) = C$ gives,

$$\frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy = 0.$$

Using the definition of the velocity potential this becomes,

$$v_x dx + v_y dy = 0.$$

Rearranging gives,

$$\frac{dy}{dx} = -\frac{v_x}{v_y}.$$

This describes lines in the xy plane with gradient $-v_x/v_y$. This is the negative reciprocal of the gradient of streamlines, and hence they are perpendicular to streamlines.

The perpendicular relationship between streamlines and potential lines fails at stagnation points, since the flow velocity vanishes and hence the gradients of streamlines and lines of constant ϕ become undefined.



EXAMPLE 33: A fluid exists for $y \geq 0$. For $y < 0$ there exists a wall. The fluid flows at normal incidence towards the wall. The flow is 2D and irrotational, and given by $\vec{v} = (v_x, v_y) = a(x, -y)$.

- (a) Determine the streamlines and sketch them (for $y \geq 0$). Highlight any stagnation point/s.
- (b) Determine the velocity potential $\phi(x, y)$ and sketch some potential lines.

6.2 Laplace's equation

For an incompressible irrotational flow, the velocity potential $\phi(\vec{x})$ satisfies Laplace's equation,

$$\nabla^2 \phi = 0 \quad (6.2)$$

We can see this by applying the incompressibility condition $\nabla \cdot \vec{v} = 0$ (Section 6.1.3) to $\vec{v} = \nabla \phi$:

$$\nabla \cdot \vec{v} = \nabla \cdot \nabla \phi = \nabla^2 \phi = 0.$$



ASIDE: Laplace's equation $\nabla^2 \phi = 0$ arises in many other contexts, such as electric fields, gravitational fields, and heat conduction. The theory of the solutions of Laplace's equation is known as *potential theory*.



ASIDE: An alternative to the velocity potential is the *stream function* $\psi(\vec{x})$. For a 2D flow this is defined via,

$$\vec{v} = (v_x, v_y) = \left(\frac{\partial \psi}{\partial y}, -\frac{\partial \psi}{\partial x} \right).$$

This definition ensures that the incompressibility condition $\nabla \cdot \vec{v} = 0$ is automatically satisfied (you might like to check this).

The stream function has the important property that lines of constant stream function are streamlines. To see this, set $\psi(x, y) = C$, a constant, and perform implicit differentiation to obtain the streamline relation, $\frac{dy}{dx} = \frac{v_y}{v_x}$.

If the flow is additionally irrotational, then ψ must satisfy Laplace's equation $\nabla^2 \psi = 0$. In other words, for incompressible irrotational flow, both the velocity potential *and* the stream function must satisfy Laplace's equation.

6.3 Superposition theorem

Laplace's equation is a *linear* equation and so satisfies the superposition theorem.

According to the **superposition theorem**, if ϕ_1 and ϕ_2 are independent solutions of the Laplace equation, then any linear sum of the form $a\phi_1 + b\phi_2$ is also a solution.

We can see this by evaluating Laplace's equation for $a\phi_1 + b\phi_2$,

$$\nabla^2(a\phi_1 + b\phi_2) = \nabla^2(a\phi_1) + \nabla^2(b\phi_2) = a\nabla^2\phi_1 + b\nabla^2\phi_2 = 0,$$

where in the final step we have used the relations that $\nabla^2\phi_1 = 0$ and $\nabla^2\phi_2 = 0$ (since ϕ_1 and ϕ_2 are independent solutions of Laplace's equation).

We call such a combined solution a *superposition*. The superposition theorem is important because it lets us construct new solutions to Laplace's equation by adding other solutions together.

6.4 Elementary flows

We next consider a group of *steady* incompressible irrotational flows, termed the *elementary flows*. These flows satisfy Laplace's equation, $\nabla^2 \phi = 0$, and we will specify their

solutions. As we will see later, the elementary flow solutions are the building blocks for constructing solutions to more complicated flows.

6.4.1 Static fluid

A **static fluid** $\vec{v} = \vec{0}$ has a uniform velocity potential, $\phi = \text{constant}$.

It is trivial to see that this ϕ recovers a zero velocity, $\nabla\phi = 0$, and that it satisfies Laplace's equation.

6.4.2 Uniform flow

A **uniform flow** $\vec{v} = (v_x, v_y)$ has a velocity potential of the form $\phi(x, y) = v_x x + v_y y$.

We can see immediately that this satisfies Laplace's equation.

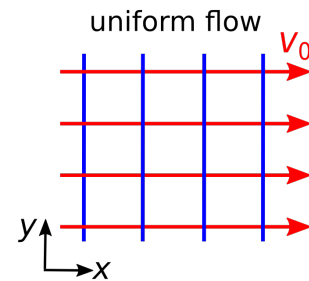
As an example, consider the flow to be in the positive x direction with constant speed v_0 , i.e. $\vec{v} = (v_x, v_y) = (v_0, 0)$. The velocity potential must satisfy,

$$\frac{\partial\phi}{\partial x} = v_0, \quad \frac{\partial\phi}{\partial y} = 0.$$

Partial integration (and dropping arbitrary constants) yields,

$$\phi = v_0 x. \quad (6.3)$$

From this we note that potential lines are lines of constant x (perpendicular to the streamlines of constant y).



Uniform flow along x , showing streamlines (red) and potential lines (blue).

6.4.3 Sources and sinks

A **source/sink** is a radial flow outwards/inwards, whose velocity potential depends only on r .

We can obtain the velocity potential for sources/sinks by seeking a solution to Laplace's equation of the form $\phi = \phi(r)$. Laplace's equation in cylindrical polar coordinates, $\nabla^2\phi = \frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) + \frac{1}{r^2} \frac{\partial^2\phi}{\partial\theta^2} = 0$, then reduces to,

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = 0.$$

Multiplying both sides by r and integrating gives $\frac{d\phi}{dr} = \frac{m}{r}$, where m is the constant of integration. Integrating again gives the velocity potential for the source/sink,

$$\phi(r) = m \ln r.$$

Note that having arrived at the potential, we ignore any additive constant of integration.

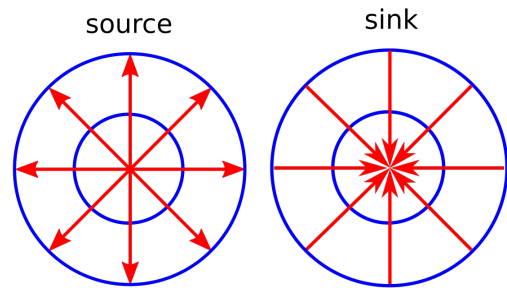
Potential lines of sources/sinks lines of constant radius, i.e. circles about the origin.

We can obtain the flow velocity for sources/sinks by evaluating $\vec{v} = \nabla\phi$ in polar coordinates,

$$\vec{v} = (v_r, v_\theta) = \nabla\phi = \left(\frac{m}{r}, 0\right).$$

Note how this is a purely radial flow, as required. Note also how it is perpendicular to the potential lines, as we would expect.

For $m = 0$ the flow is static. For $m > 0$ we have an outward flow (*source*), while for $m < 0$ we have an inward flow (*sink*). The origin is a *singular point* of the flow: at this point, the magnitude of the velocity is infinite and the flow is in all directions.



Source/sink flows, with streamlines (red) and potential lines (blue).

6.4.4 Free vortex

A **free vortex** is a circulating flow whose velocity potential depends only on θ .



EXAMPLE 34:

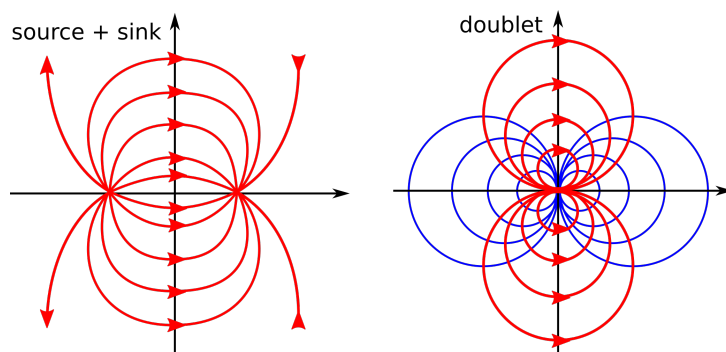
- Derive the velocity potential of the free vortex by seeking a solution of Laplace's $\nabla^2\phi = 0$ of the form $\phi = \phi(\theta)$.
- Determine the flow velocity of this solution.
- Sketch some streamlines and potential lines of this flow.
- Obtain an expression for how the pressure varies with radius, taking $P(r = \infty) = P_0$.

6.4.5 Doublet

Using the superposition theorem we will construct another important solution to Laplace's equation - the *doublet* - by combining a source and a sink.

A **doublet** is a superposition of a source and sink, which are infinitesimally close.

First consider combining a source at $(x, y) = (-b, 0)$ and a sink of equal magnitude at $(x, y) = (+b, 0)$, where b is a constant; the streamlines (shown below) are emitted by the source and get drawn in to the sink. Now reduce b until the source and sink are infinitesimally close (but not quite overlapping). This creates the *doublet*, depicted below.



Left: Streamlines of a flow combining a source and sink. Streamlines (red) and potential lines (blue) of the doublet solution.

The velocity potential for the doublet is,

$$\phi = \frac{a \cos \theta}{r}. \quad (6.4)$$

Potential lines are circles in a figure-of-eight pattern, lying along the x -axis. Streamlines have the same shape but are rotated by 90° .



EXAMPLE 35: Consider a 2D potential flow consisting of a source of strength m at $(x, y) = (-b, 0)$ and a sink of strength $-m$ at $(x, y) = (b, 0)$.

- Construct the velocity potential $\phi(x, y)$ for this flow.
- Show that, as $b \rightarrow 0$, we obtain the doublet solution $\phi(r, \theta) = a \cos \theta / r$.
Hint: $\ln \sqrt{(x \pm b)^2 + y^2} \approx \ln(r) \pm bx/r^2$ as $b \rightarrow 0$.
- Show that potential lines are circles whose centre is shifted from the origin along the x axis.

6.5 Solving for general flow scenarios

We have solved the simplest cases of incompressible irrotational flow, such as uniform flows and sources/sinks. By “solving” we mean that we found the corresponding solution ϕ to Laplace’s equation. From this we can go on to calculate useful quantities like the flow velocity (through $\vec{v} = \nabla \phi$) and the pressure distribution (using Bernoulli’s theorem).

But how do we solve a more complicated flow scenario, for example, for a fluid incident upon an obstacle? We must find the solution ϕ which satisfies the boundary conditions for the flow scenario. We can construct this solution by using the elementary solutions as building blocks (and thanks to the superposition theorem, we know that such a solution will surely satisfy Laplace’s equation). A steady flow solution can be constructed of elementary solutions which are fixed in space, while an unsteady flow can be constructed of elementary solutions which move around in time.

We will not aim to obtain such solutions from scratch; this is usually very challenging to do. Instead we will focus on constructing and examining some known solutions, focussing on steady flows for simplicity.

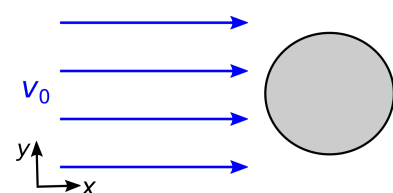


EXAMPLE 36:

- Construct a velocity potential $\phi(r, \theta)$ which combines uniform flow in the positive x direction of speed v_0 with a source at the origin of strength m ($m > 0$).
- Obtain the flow velocity $\vec{v}(r, \theta)$.
- Obtain the flow speed $v = |\vec{v}|$. What value does it approach for $r \rightarrow \infty$?
- Show that there exists one stagnation point, and provide its coordinates in both polar coordinates and Cartesian coordinates.

6.6 Steady flow past obstacles

We will consider the general scenario shown to the right: the flow is incident in the positive x direction upon a stationary object.



2D flow upon an obstacle.

6.6.1 Boundary conditions

There are two boundary conditions to this scenario:

Far-field condition: Far away from the obstacle (the “far-field”), the flow recovers to the bulk flow $\vec{v} = (v_x, v_y) = (v_0, 0)$.

Surface condition: Fluid cannot flow through the surface of the object S . This requires that the component of the flow normal to the surface S must be zero according to,

$$\vec{v} \cdot \hat{n} = 0 \quad \text{on } S. \quad (6.5)$$

6.6.2 Virtual objects

The condition for the object boundary, Eq. (6.5), also holds true for any streamline: the flow has zero component normal to a streamline (since a streamline by definition is tangential to the flow). Thus we can think of the object boundary as being a streamline. Or, vice versa, we can treat any streamline of the flow as if it were the boundary of a solid object!

This means that we do not need to explicitly include the object in our solution as a region which is free of fluid: instead we can consider the whole domain as containing fluid, providing it possesses a streamline (the “dividing streamline”) which permanently divides the flow into an inner region (the object) and an outer region (the flow around it).

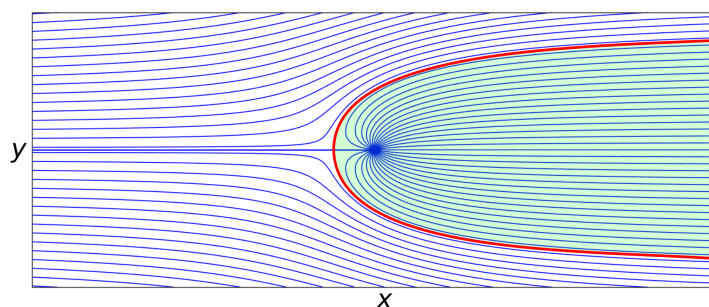
6.6.3 The Rankine half-body

Consider again the steady flow in Example 36: uniform flow along x (speed v_0) plus a source (strength m) at the origin. The streamlines are shown below. The red streamline may be treated as the boundary of the blunt-shaped green region (which might represent, for example, the nose of an aircraft or a bullet). We call this shape the **Rankine half-body**. It is referred to as a “half body” since it is open at one end.

The boundary of the object/dividing streamline is most conveniently expressed in polar coordinates as the curve*,

$$r(\theta) = \frac{m(\pi - \theta)}{v_0 \sin \theta}.$$

The one stagnation point we found in Example 36 lies at the tip of this object: it is formed when the uniform flow to the right cancels out with the flow to the left emanating from the source. From Example 36 we also know that the flow velocity asymptotes to $\vec{v} = (v_x, v_y) = (v_0, 0)$, the incident flow velocity, as we move away from the object.



Streamlines (blue lines) past the Rankine half-body (green region). The boundary/dividing streamline is shown in red.

*The derivation of this curve is beyond the scope of this module.

6.6.4 Flow past a cylinder

Next we consider the flow past the circular cross-section of a cylindrical object. The circle has radius R and is centred at the origin. Our solution ϕ must (i) recover the uniform flow $\vec{v} = (v_x, v_y) = (v_0, 0)$ in the far-field, and (ii) possess a streamline at $r = R$ to mimic the circle's boundary.

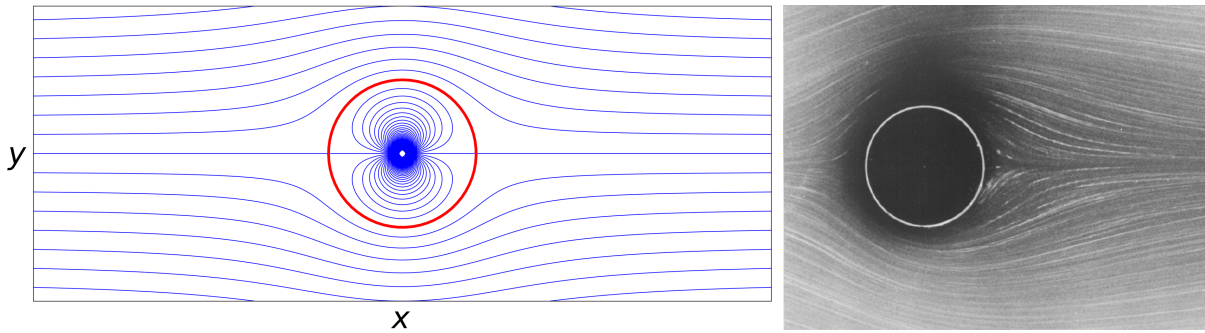
Such a solution (in polar coordinates) can be obtained by combining the uniform flow $\phi = v_0 x$ with the doublet solution, $\phi = v_0 R^2 \cos \theta / r$, giving

$$\phi = v_0 \cos \theta \left(r + \frac{R^2}{r} \right).$$

Calculating $\nabla \phi$ yields the flow velocity,

$$\vec{v} = (v_r, v_\theta) = \left(v_0 \left(1 - \frac{R^2}{r^2} \right) \cos \theta, -v_0 \left(1 + \frac{R^2}{r^2} \right) \sin \theta \right). \quad (6.6)$$

The streamlines for this flow are shown below. It gives good agreement with experimental images of flow past a cylinder.



Left: Streamlines (blue lines) past a cross-section of a cylinder (boundary/dividing streamline shown in red). Right: An experiment showing the streamlines past a cylindrical object.

We can make several observations and predictions from the above flow velocity:

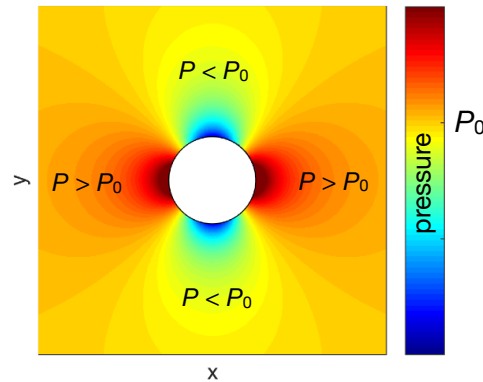
1. The radial velocity component v_r clearly vanishes at $r = R$, thus it satisfies the surface condition of Eq. 6.5.
2. There are two stagnation points, at $(r, \theta) = (R, 0)$ and (R, π) .
3. On the boundary ($r = R$), the flow is fastest at the “poles” of the circle, $(r, \theta) = (R, \pm\pi/2)$, where it has the speed $2v_0$.
4. We can use the irrotational Bernoulli's theorem to state the pressure everywhere in the flow. Equating the Bernoulli function in the far-field (pressure P_0 , flow speed v_0) and at an arbitrary point in the flow (pressure P , flow speed $v = |\vec{v}|$) gives,

$$P_0 + \frac{1}{2}\rho v_0^2 = P(\vec{x}) + \frac{1}{2}\rho v(\vec{x})^2. \quad (6.7)$$

Rearranging for the pressure $P(\vec{x})$ and introducing the flow velocity Eq (6.6) gives,

$$P(\vec{x}) = \frac{1}{2}\rho (v_0^2 - v(\vec{x})^2) + P_0, \quad (6.8)$$

$$= \frac{1}{2}\rho v_0^2 \left[\frac{2R^2}{r^2} \cos 2\theta - \frac{R^4}{r^4} \right] + P_0. \quad (6.9)$$

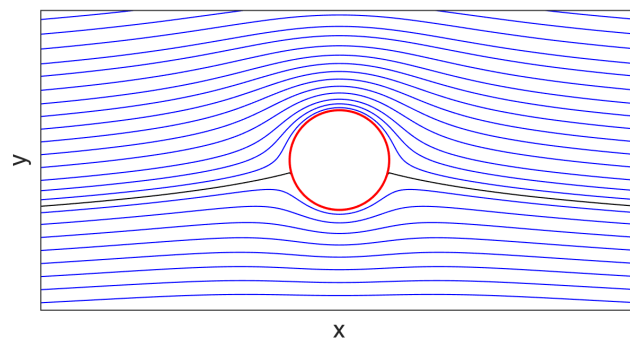


We have used the trig identity $\cos^2 \theta - \sin^2 \theta = \cos 2\theta$ to write the result in a more compact form. This pressure distribution is plotted below.

5. The maximum pressure occurs at the stagnation points and the minimum pressure occurs at the poles.
6. The pressure is symmetric about $x = 0$ and about $y = 0$. In particular, the latter symmetry means that there is no vertical pressure gradient and hence no vertical pressure force on the cylinder. In other words, the cylinder won't fly!



ASIDE: For the above flow scenario, there is no net pressure force on the object. However, now consider adding a free vortex at the origin to our solution. The streamlines are shown below.



The symmetry about $y = 0$ is now broken. The fluid travels faster above the cylinder than below it. By the Bernoulli effect, this means that the pressure is lower above the cylinder than below. The difference in pressure creates a pressure force directed upwards. This would enable the cylinder to defy gravity, and fly!

The same principle - an asymmetric flow pattern around the object - generates lift on wings and is central to flight. The wing must be shaped in such a way that favours an asymmetric flow to be produced and be maximised, which is why wings are shaped like aerofoils and not cylinders. Further study of wings, lift and flight will be covered in the stage 3 module MAS3803 Advanced Fluid Dynamics.

A Appendices

A.1 Integrating/differentiating a vector field in polar coordinates

Consider the 2D plane and the Cartesian coordinate system. The unit vector along x , $\hat{x} = (1, 0)$, is the same at any location in the plane. The same is true for the unit vector along y , $\hat{y} = (0, 1)$. This means that when we integrate or differentiate them with respect to position, we can treat them as a constant.

So, as an example of differentiation, consider $\frac{d}{dx}(7\hat{x}) = 0$. As an example of integrating, consider $\int_0^1 7\hat{x} dx = 7\hat{x} \int_0^1 dx = 7\hat{x}$. The above property makes vectors easy to integrate and differentiate in Cartesian coordinates.

Unfortunately, this does not hold in polar coordinates. This is because the radial unit vector $\hat{r}$ and the azimuthal unit vector $\hat{\theta}$ depend on position. More specifically, they both depend on the polar angle θ (they do not depend on r or z). Picture the radial unit vector pointing outwards from the origin: when $\theta = 0$, $\hat{r}$ points along x and when $\theta = \pi/2$, $\hat{r}$ points along y . Clearly there is a dependence on θ . The azimuthal unit vector $\hat{\theta}$ is always perpendicular to $\hat{r}$ (in the positive θ direction), and so this unit vector must also depend on θ .

This means that when we differentiate or integrate these unit vectors, we must take into account that they depend on θ . How do they depend on θ ? By trigonometry we can write $\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$ and $\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$.

So how do we evaluate the derivative, say, $\frac{d}{d\theta}(7\hat{r})$? We introduce the θ -dependence of $\hat{r}$ before doing the differentiation:

$$\frac{d}{d\theta}(7\hat{r}) = \frac{d}{d\theta}(7 \cos \theta \hat{x} + 7 \sin \theta \hat{y}) \quad (\text{A.1})$$

$$= -7 \sin \theta \hat{x} + 7 \cos \theta \hat{y}. \quad (\text{A.2})$$

If we wanted to, we could write this answer back in terms of the polar unit vectors as $7\hat{\theta}$.

What about the integral, say, $\int_0^{2\pi} 7\hat{r} d\theta$? Similarly, we must introduce the θ -dependence before integrating:

$$\int_0^{2\pi} 7\hat{r} d\theta = \int_0^{2\pi} (7 \cos \theta \hat{x} + 7 \sin \theta \hat{y}) d\theta \quad (\text{A.3})$$

$$= [7 \sin \theta \hat{x} - 7 \cos \theta \hat{y}]_0^{2\pi} \quad (\text{A.4})$$

$$= (0 - 7\hat{y}) - (0 - 7\hat{y}) \quad (\text{A.5})$$

$$= 0. \quad (\text{A.6})$$

We would use a similar method if differentiating/integrating a function involving $\hat{\theta}$. For example,

$$\frac{d}{d\theta}(7\hat{\theta}) = \frac{d}{d\theta}(-7 \sin \theta \hat{x} + 7 \cos \theta \hat{y}) \quad (\text{A.7})$$

$$= 7 \cos \theta \hat{x} + 7 \sin \theta \hat{y}. \quad (\text{A.8})$$

And for example:

$$\int_0^{2\pi} 7\hat{\theta} d\theta = \int_0^{2\pi} (-7 \sin \theta \hat{x} + 7 \cos \theta \hat{y}) d\theta \quad (\text{A.9})$$

$$= [7 \cos \theta \hat{x} + 7 \sin \theta \hat{y}]_0^{2\pi} \quad (\text{A.10})$$

$$= (7\hat{x} + 0) - (7\hat{x} + 0) \quad (\text{A.11})$$

$$= 0. \quad (\text{A.12})$$

With this method we can do any integration/differentiation of vectors in polar (or cylindrical polar) coordinates.

A.2 Fluid Acceleration in Cylindrical Polar Coordinates

The fluid acceleration $\vec{a} = \frac{D\vec{v}}{Dt}$ is more complicated in cylindrical polar coordinates than in Cartesian coordinates. Let's see why and derive it's form.

First let's derive some useful general relations we will need later. We can write the radial and azimuthal unit vectors as,

$$\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}, \quad \hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}.$$

It follows that,

$$\hat{r} = -\frac{\partial \hat{\theta}}{\partial \theta}, \quad \hat{\theta} = \frac{\partial \hat{r}}{\partial \theta}. \quad (\text{A.13})$$

We are considering a velocity expressed in cylindrical polar coordinates as $\vec{v} = v_r \hat{r} + v_\theta \hat{\theta} + v_z \hat{z} = (v_r, v_\theta, v_z)$. Now we start evaluating the fluid acceleration from its definition,

$$\frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v}.$$

Since in cylindrical polar coordinates the gradient operator is defined as $\nabla = \left(\frac{\partial}{\partial r}, \frac{1}{r} \frac{\partial}{\partial \theta}, \frac{\partial}{\partial z} \right)$ we can write the material derivative as,

$$\frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + v_r \frac{\partial \vec{v}}{\partial r} + \frac{v_\theta}{r} \frac{\partial \vec{v}}{\partial \theta} + v_z \frac{\partial \vec{v}}{\partial z}. \quad (\text{A.14})$$

Consider the time derivative term in Eq. (A.14) and expanding it out using the chain rule gives,

$$\frac{\partial \vec{v}}{\partial t} = \frac{\partial(v_r \hat{r})}{\partial t} + \frac{\partial(v_\theta \hat{\theta})}{\partial t} + \frac{\partial(v_z \hat{z})}{\partial t} = v_r \frac{\partial \hat{r}}{\partial t} + \hat{r} \frac{\partial v_r}{\partial t} + v_\theta \frac{\partial \hat{\theta}}{\partial t} + \hat{\theta} \frac{\partial v_\theta}{\partial t} + v_z \frac{\partial \hat{z}}{\partial t} + \hat{z} \frac{\partial v_z}{\partial t}.$$

We are computing the partial derivative with time at a fixed position, hence the unit vectors are fixed and $\frac{\partial \hat{r}}{\partial t} = \frac{\partial \hat{\theta}}{\partial t} = \frac{\partial \hat{z}}{\partial t} = 0$, such that $\frac{\partial \vec{v}}{\partial t}$ simplifies to,

$$\frac{\partial \vec{v}}{\partial t} = \frac{\partial v_r}{\partial t} \hat{r} + \frac{\partial v_\theta}{\partial t} \hat{\theta} + \frac{\partial v_z}{\partial t} \hat{z}.$$

Now we consider the remaining terms in Eq. (A.14). These terms involve the derivative of $\vec{v}$ with respect to r , θ and z . Here is where the real complication comes in. Two of the

components of $\vec{v}$ are along the $\hat{r}$ and $\hat{\theta}$ unit vectors. The unit vectors themselves depend on position, and this complicates taking their derivative. We must carefully expand out the derivatives using the chain rule.

First let's consider the $v_r \frac{\partial \vec{v}}{\partial r}$ term in Eq. (A.14). Expanding it out and using the chain rule,

$$\begin{aligned} v_r \frac{\partial \vec{v}}{\partial r} &= v_r \left(\frac{\partial(v_r \hat{r})}{\partial r} + \frac{\partial(v_\theta \hat{\theta})}{\partial r} + \frac{\partial(v_z \hat{z})}{\partial r} \right) = v_r \left(\cancel{v_r} \frac{\partial \hat{r}}{\partial r} + \hat{r} \frac{\partial v_r}{\partial r} + \cancel{v_\theta} \frac{\partial \hat{\theta}}{\partial r} + \hat{\theta} \frac{\partial v_\theta}{\partial r} + \cancel{v_z} \frac{\partial \hat{z}}{\partial r} + \hat{z} \frac{\partial v_z}{\partial r} \right) \\ &= v_r \left(\frac{\partial v_r}{\partial r} \hat{r} + \frac{\partial v_\theta}{\partial r} \hat{\theta} + \frac{\partial v_z}{\partial r} \hat{z} \right) \end{aligned} \quad (\text{A.15})$$

Similarly, let's consider the $\frac{v_\theta}{r} \frac{\partial \vec{v}}{\partial \theta}$ term in Eq. (A.14),

$$\begin{aligned} \frac{v_\theta}{r} \frac{\partial \vec{v}}{\partial \theta} &= \frac{v_\theta}{r} \left(\frac{\partial(v_r \hat{r})}{\partial \theta} + \frac{\partial(v_\theta \hat{\theta})}{\partial \theta} + \frac{\partial(v_z \hat{z})}{\partial \theta} \right) = \frac{v_\theta}{r} \left(v_r \frac{\partial \hat{r}}{\partial \theta} + \hat{r} \frac{\partial v_r}{\partial \theta} + v_\theta \frac{\partial \hat{\theta}}{\partial \theta} + \hat{\theta} \frac{\partial v_\theta}{\partial \theta} + \cancel{v_z} \frac{\partial \hat{z}}{\partial \theta} + \hat{z} \frac{\partial v_z}{\partial \theta} \right) \\ &= \frac{v_\theta}{r} \left(\left(\frac{\partial v_r}{\partial \theta} - v_\theta \right) \hat{r} + \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right) \hat{\theta} + \frac{\partial v_z}{\partial \theta} \hat{z} \right), \end{aligned} \quad (\text{A.16})$$

where we have used the relations that $\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$ and $\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$.

And finally for the $v_z \frac{\partial \vec{v}}{\partial z}$ term in Eq. (A.14),

$$\begin{aligned} v_z \frac{\partial \vec{v}}{\partial z} &= v_z \left(\frac{\partial(v_r \hat{r})}{\partial z} + \frac{\partial(v_\theta \hat{\theta})}{\partial z} + \frac{\partial(v_z \hat{z})}{\partial z} \right) = v_z \left(\cancel{v_r} \frac{\partial \hat{r}}{\partial z} + \hat{r} \frac{\partial v_r}{\partial z} + \cancel{v_\theta} \frac{\partial \hat{\theta}}{\partial z} + \hat{\theta} \frac{\partial v_\theta}{\partial z} + \cancel{v_z} \frac{\partial \hat{z}}{\partial z} + \hat{z} \frac{\partial v_z}{\partial z} \right) \\ &= v_z \left(\frac{\partial v_r}{\partial z} \hat{r} + \frac{\partial v_\theta}{\partial z} \hat{\theta} + \frac{\partial v_z}{\partial z} \hat{z} \right) \end{aligned} \quad (\text{A.17})$$

Putting Eqs. (A.15, A.16 and A.17) together, the material derivative in cylindrical polar coordinates is,

$$\begin{aligned} \frac{D\vec{v}}{Dt} &= \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z \frac{\partial v_r}{\partial z} \right) \hat{r} \\ &\quad + \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta v_r}{r} + v_z \frac{\partial v_\theta}{\partial z} \right) \hat{\theta} \\ &\quad + \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) \hat{z} \end{aligned}$$

A.3 Gauss' theorem

Let V be a volume surrounded by the closed surface S . Then for a some general differentiable vector function $\vec{F} = \vec{F}(\vec{x})$ whose derivatives are continuous,

$$\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \cdot \vec{F} dV,$$

where $\hat{n}$ is the outward-pointing unit normal.

An alternative statement of Gauss' theorem is for a scalar function $\phi = \phi(\vec{x})$. Providing $\nabla\phi$ is continuous,

$$\iint_S \phi \hat{n} dS = \iiint_V \nabla\phi dV.$$

A.4 Stokes' theorem

Let C be a closed curve in 3D space, and let S be any surface bounded by C . Then, for some general differentiable vector function $\vec{F} = \vec{F}(\vec{x})$,

$$\oint_C \vec{F} \cdot d\vec{\ell} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS,$$

where $\hat{n}$ is the unit vector normal to S .

A.5 Partial Integration

Consider a function $h(x, y, z)$ which satisfies some partial differential equations of the form,

$$\frac{\partial h}{\partial x} = a(x, y, z), \quad \frac{\partial h}{\partial y} = b(x, y, z), \quad \frac{\partial h}{\partial z} = c(x, y, z).$$

We can construct the solution for $h(x, y, z)$ which satisfies all three equations by the method of "partial integration". This name refers to the *integration of partial derivatives*.

For example, consider three partial differential equations,

$$\frac{\partial h}{\partial x} = x^2, \quad \frac{\partial h}{\partial y} = 3y, \quad \frac{\partial h}{\partial z} = 1.$$

Integrating each, and taking care with the "partial" nature of the derivatives, gives,

$$\begin{cases} h(x, y, z) = \int x^2 \partial x = \frac{x^3}{3} + A(y, z) + c_1, \\ h(x, y, z) = \int 3y \partial y = \frac{3y^2}{2} + B(x, z) + c_2, \\ h(x, y, z) = \int 1 \partial z = z + C(x, y) + c_3. \end{cases}$$

We introduce arbitrary functions, $A(y, z)$, $B(x, z)$ and $C(x, y)$, since the integrals can have such functions and still recover the original differential equations. This is in addition to the usual arbitrary constants of integration c_1 , c_2 and c_3 .

We construct a general solution for $h(x, y, z)$ as the *simplest function which satisfies all of the above equations*, which in this case is,

$$h(x, y, z) = \frac{x^3}{3} + \frac{3y^2}{2} + z + c,$$

where c is a combined arbitrary constant.

A.6 Vector identities

Let $\vec{A}$, $\vec{B}$ be vector fields, and f be a scalar field.

$$\begin{aligned}\nabla(\vec{A} \cdot \vec{B}) &= \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A}) + (\vec{A} \cdot \nabla) \vec{B} + (\vec{B} \cdot \nabla) \vec{A}, \\ \nabla \cdot (f \vec{A}) &= f \nabla \cdot \vec{A} + \vec{A} \cdot \nabla f, \quad \nabla \cdot (\nabla \times \vec{A}) = 0, \\ \nabla^2 \vec{A} &= \nabla(\nabla \cdot \vec{A}) - \nabla \times (\nabla \times \vec{A}).\end{aligned}$$

A.7 Vector derivatives in cylindrical polar coordinates

Consider cylindrical coordinates (r, θ, z) . Let f be a scalar field and $\vec{v} = (v_r, v_\theta, v_z)$ be a vector field. The form of the gradient, divergence, curl and Laplacian are

$$\begin{aligned}\nabla f &= \left(\frac{\partial f}{\partial r}, \frac{1}{r} \frac{\partial f}{\partial \theta}, \frac{\partial f}{\partial z} \right) \\ \nabla \cdot \vec{v} &= \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \\ \nabla \times \vec{v} &= \left(\frac{1}{r} \frac{\partial v_z}{\partial \theta} - \frac{\partial v_\theta}{\partial z}, \frac{\partial v_r}{\partial z} - \frac{\partial v_z}{\partial r}, \frac{1}{r} \frac{\partial (r v_\theta)}{\partial r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right) \\ \nabla^2 f &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}.\end{aligned}$$